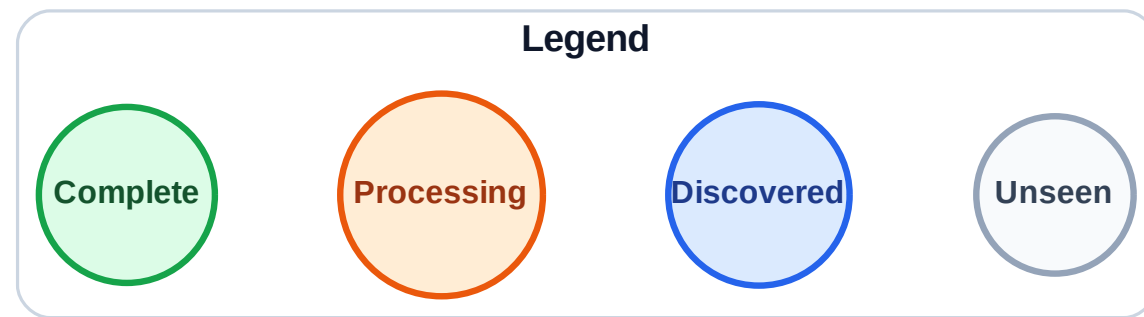
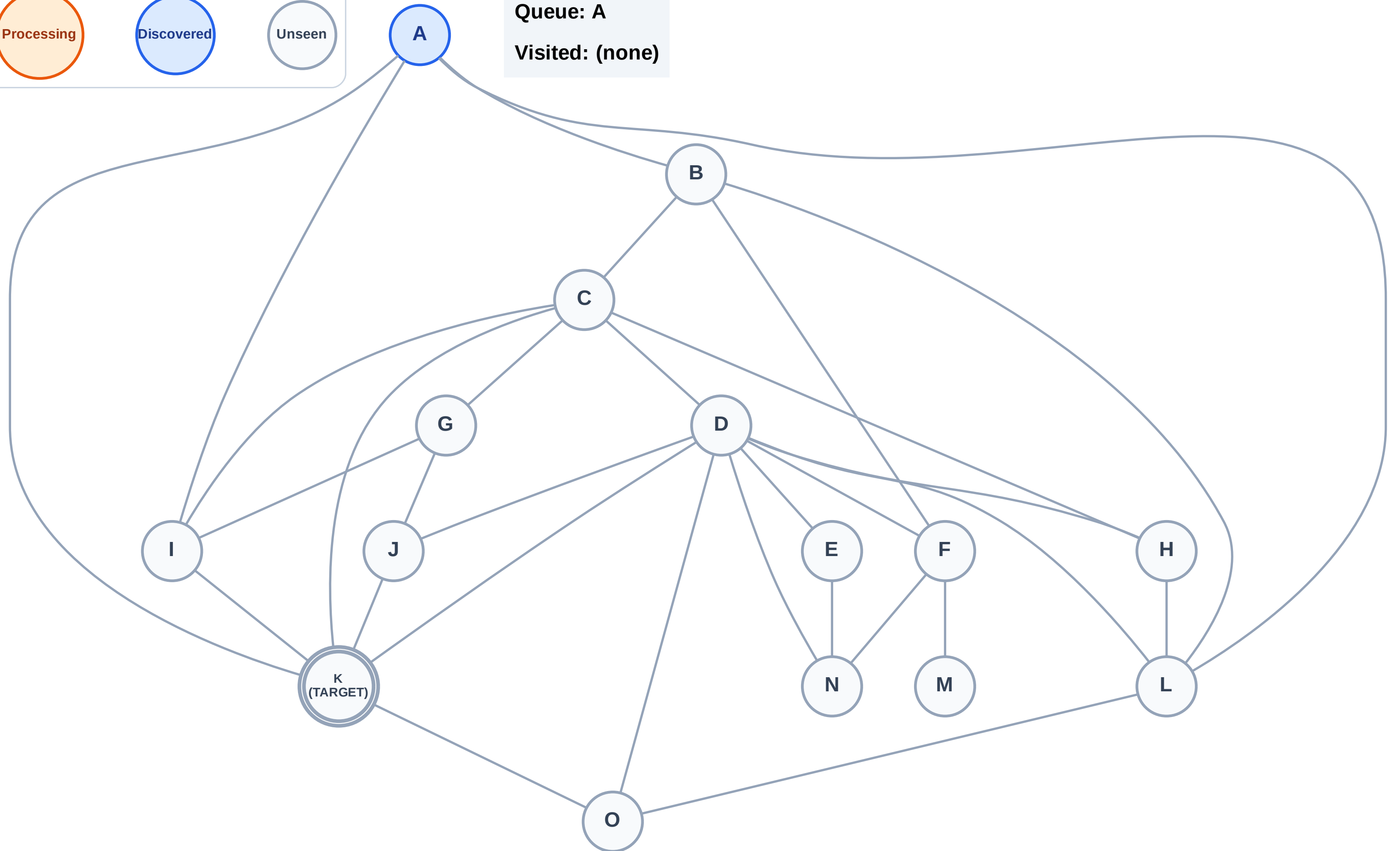


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

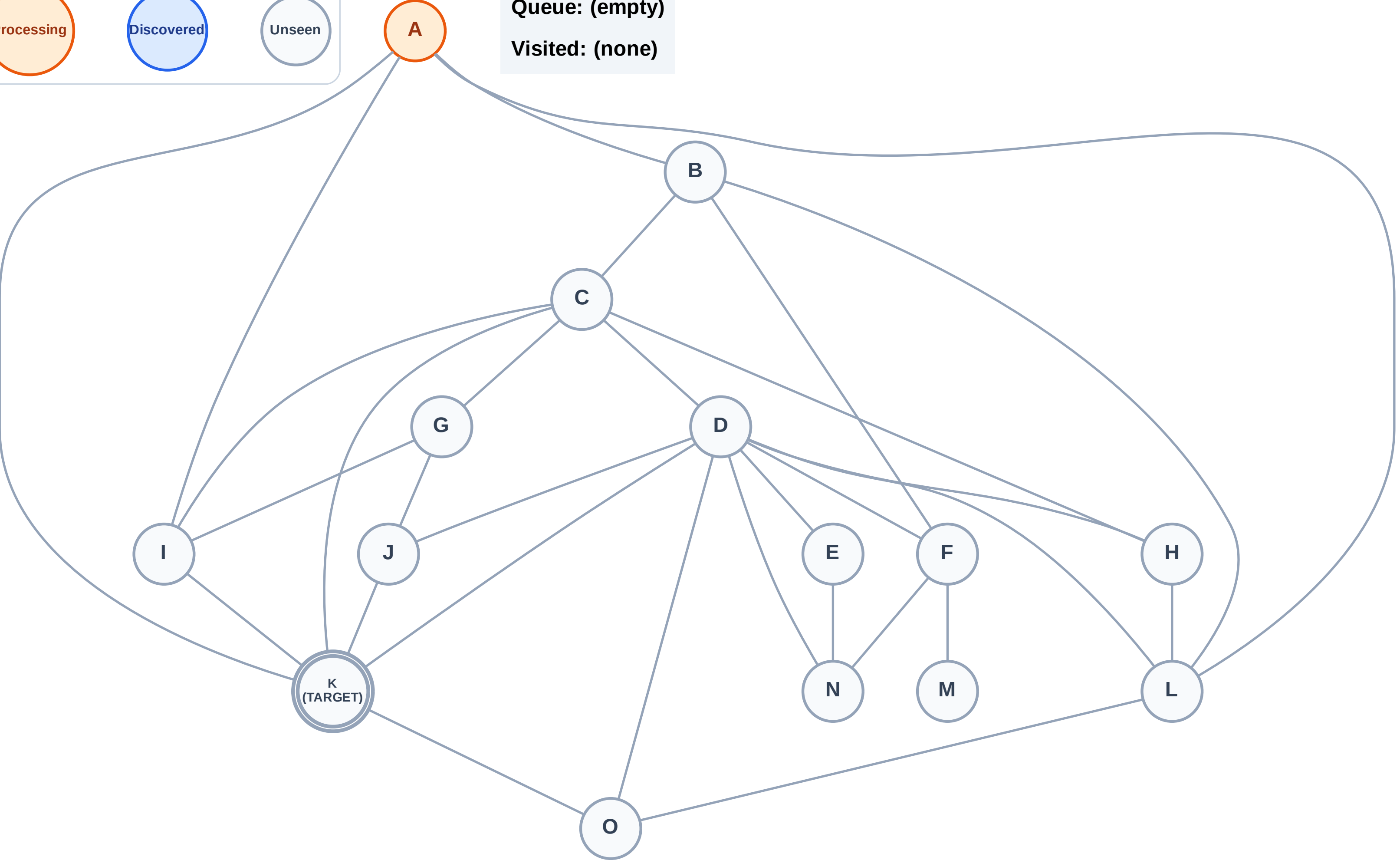
Processing

Discovered

Unseen

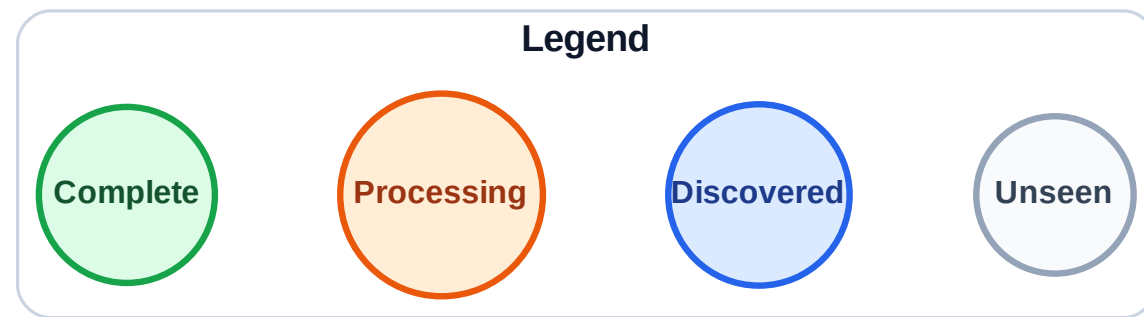
Queue: (empty)

Visited: (none)



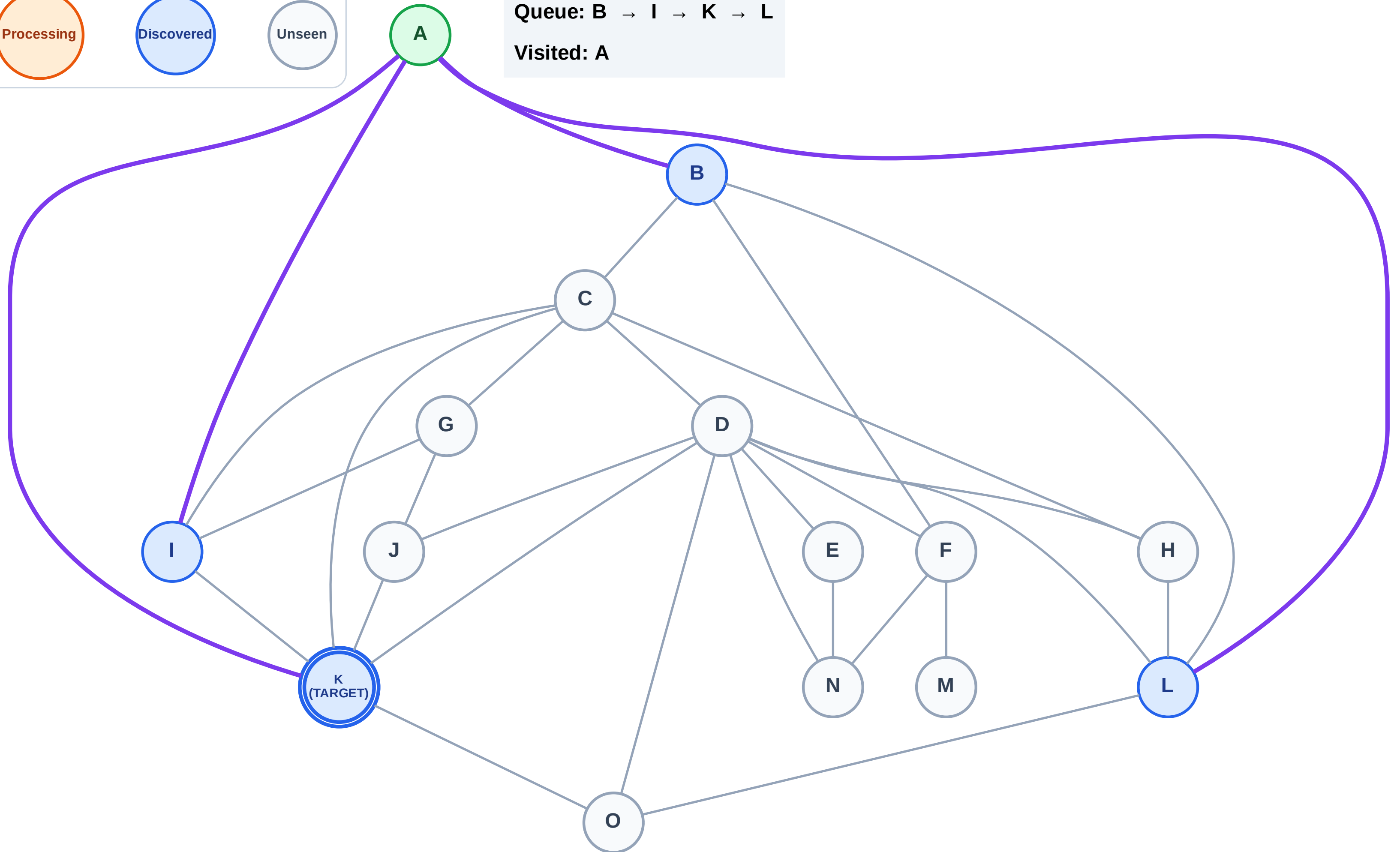
BFS Traversal

Step 3: Discover B, I, K, L from A



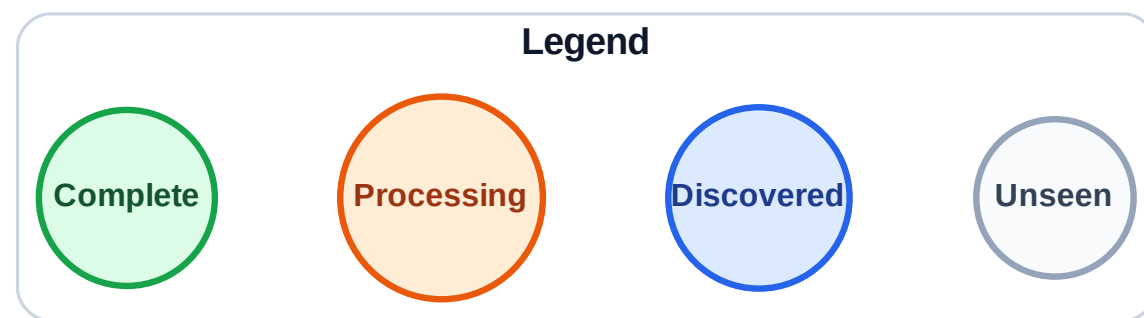
Queue: B → I → K → L

Visited: A



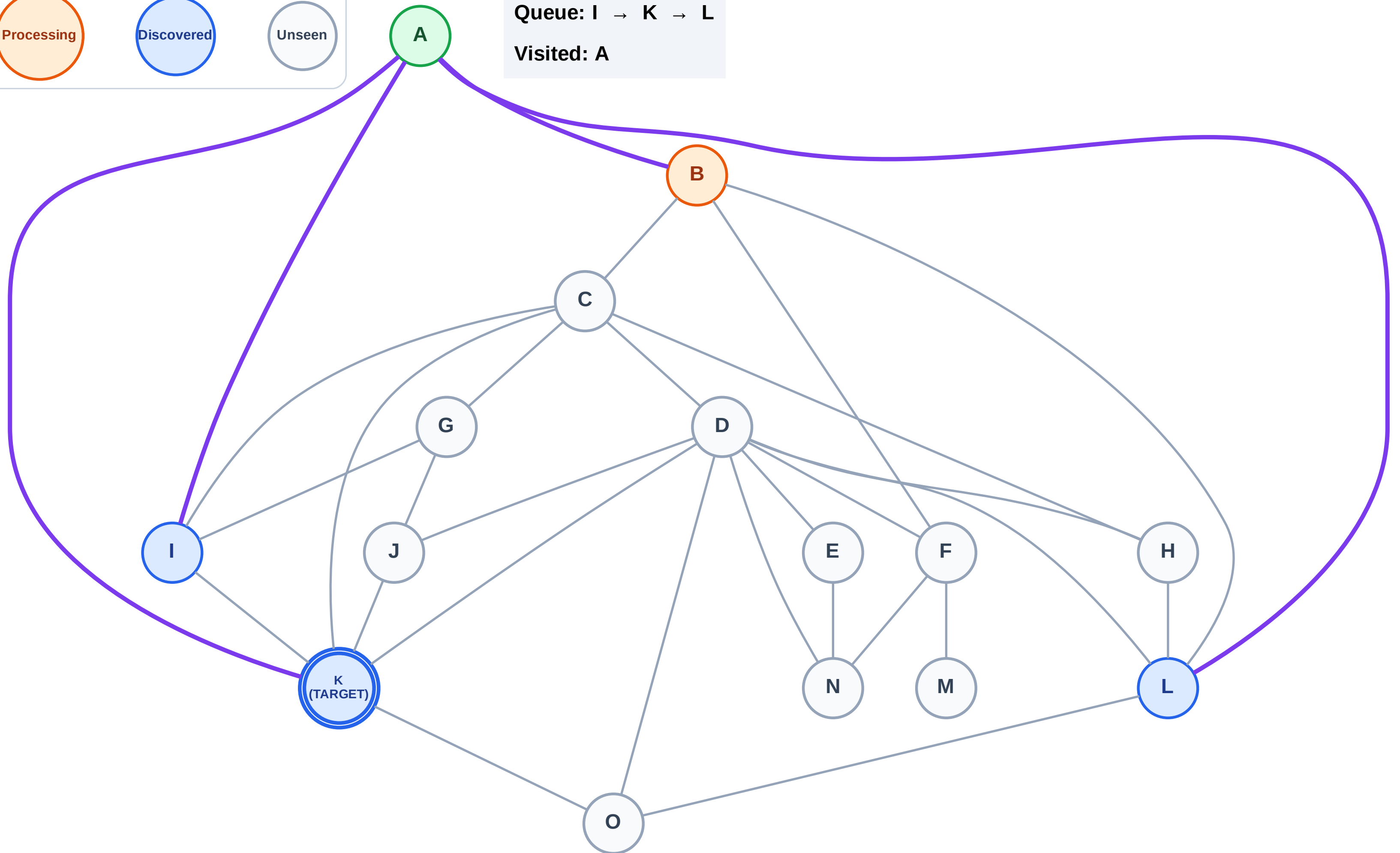
BFS Traversal

Step 4: Process node B



Queue: I → K → L

Visited: A



BFS Traversal

Step 5: Discover C, F from B

Queue: I → K → L → C → F
Visited: A, B

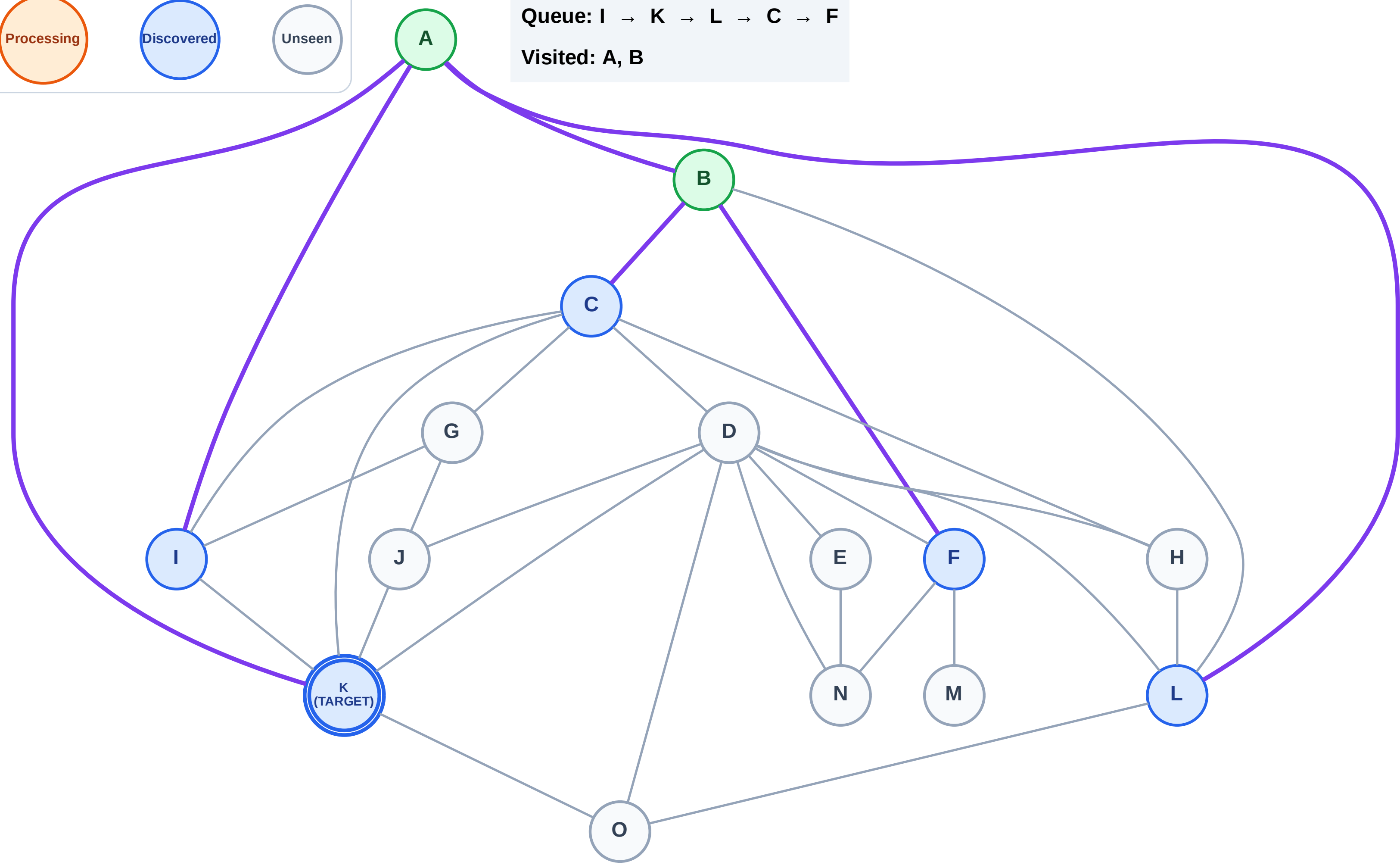
Legend

Complete

Processing

Discovered

Unseen



BFS Traversal

Step 6: Process node I

Legend

Complete

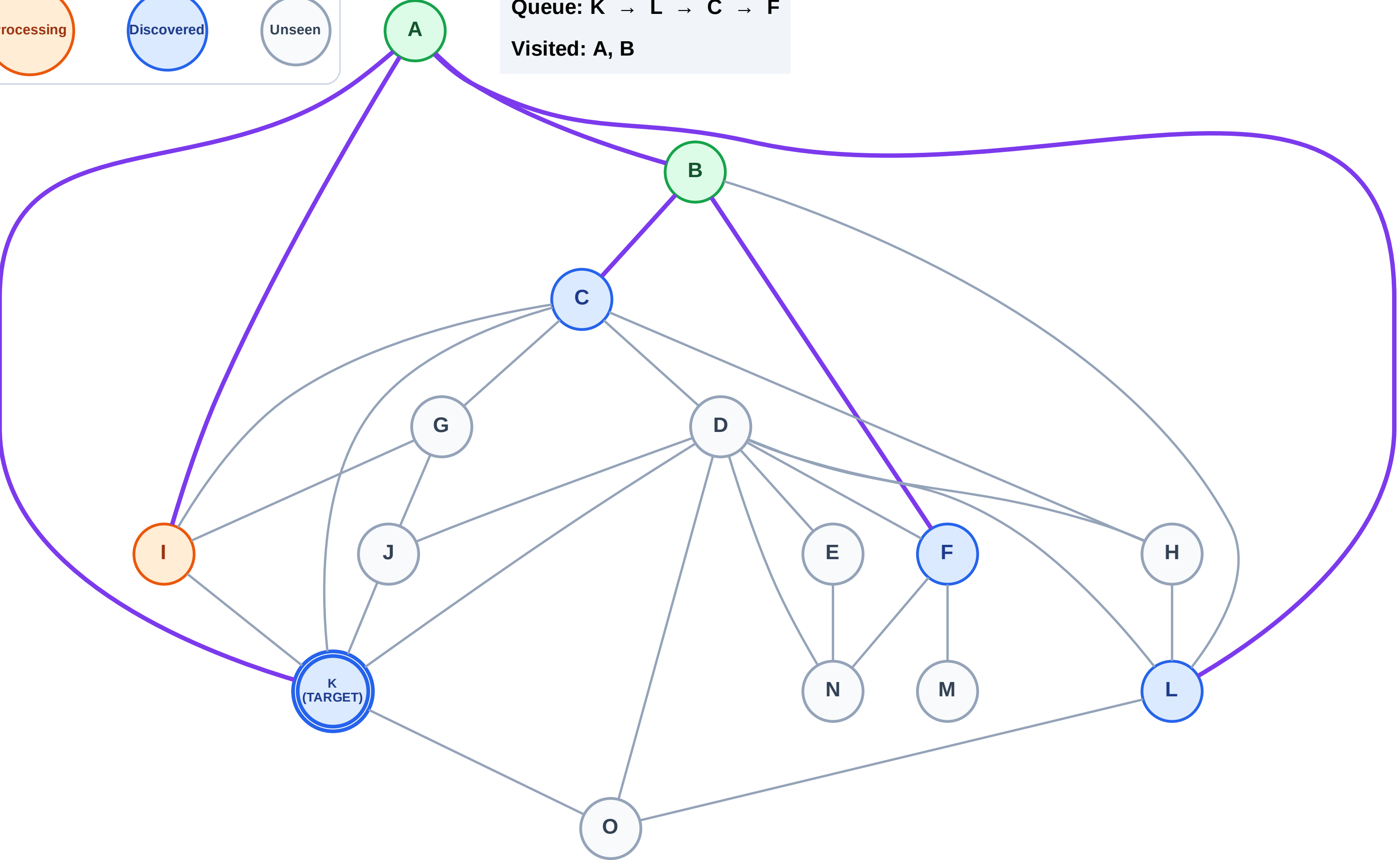
Processing

Discovered

Unseen

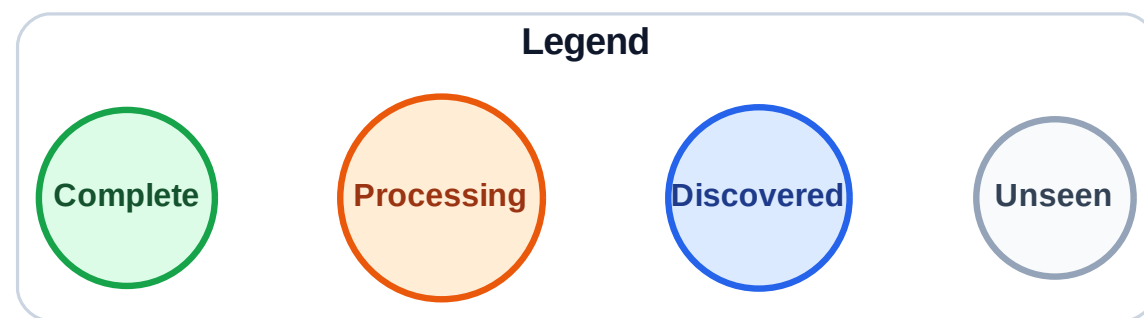
Queue: K → L → C → F

Visited: A, B



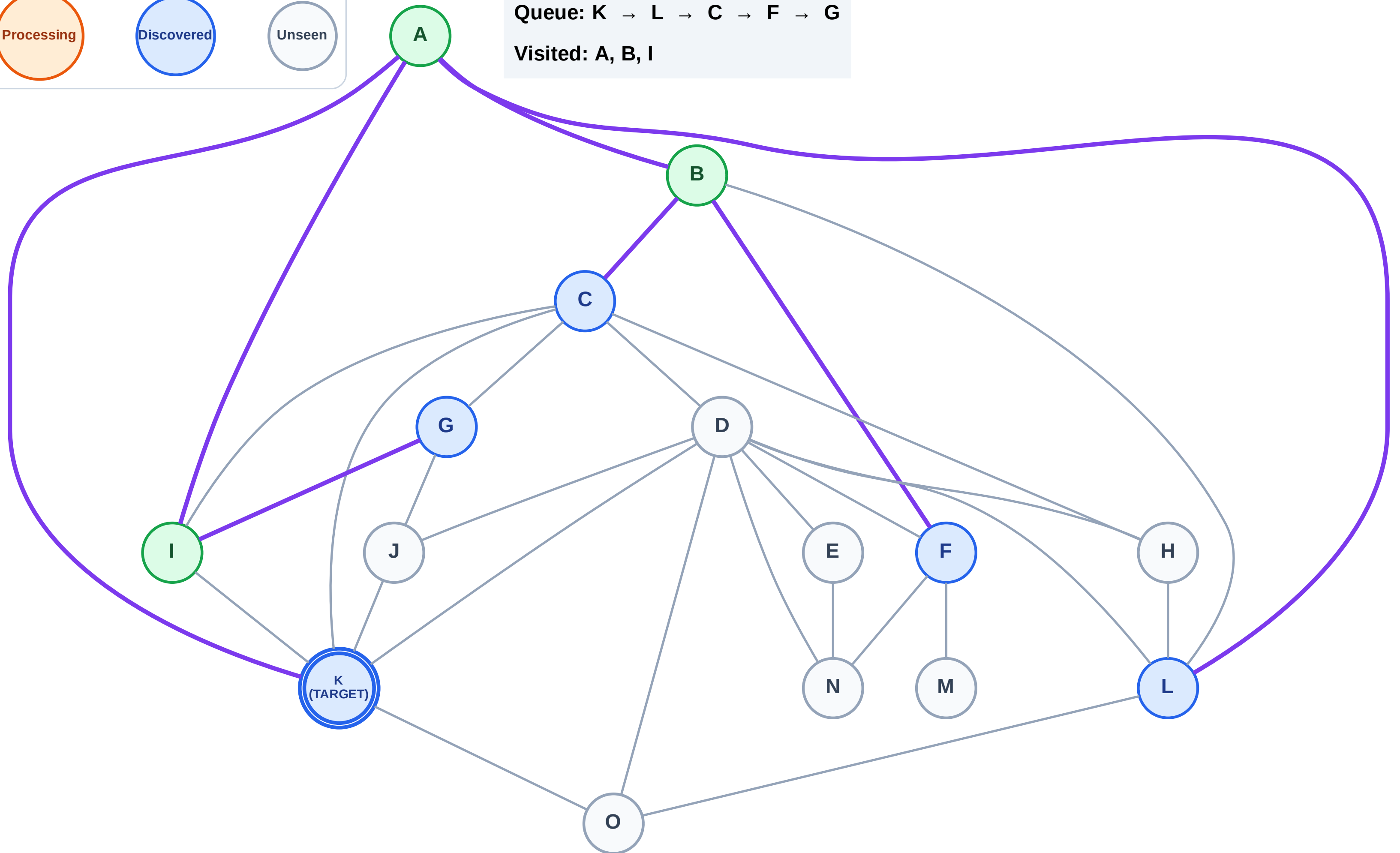
BFS Traversal

Step 7: Discover G from I



Queue: K → L → C → F → G

Visited: A, B, I



BFS Traversal

Step 8: Process node K

Legend

Complete

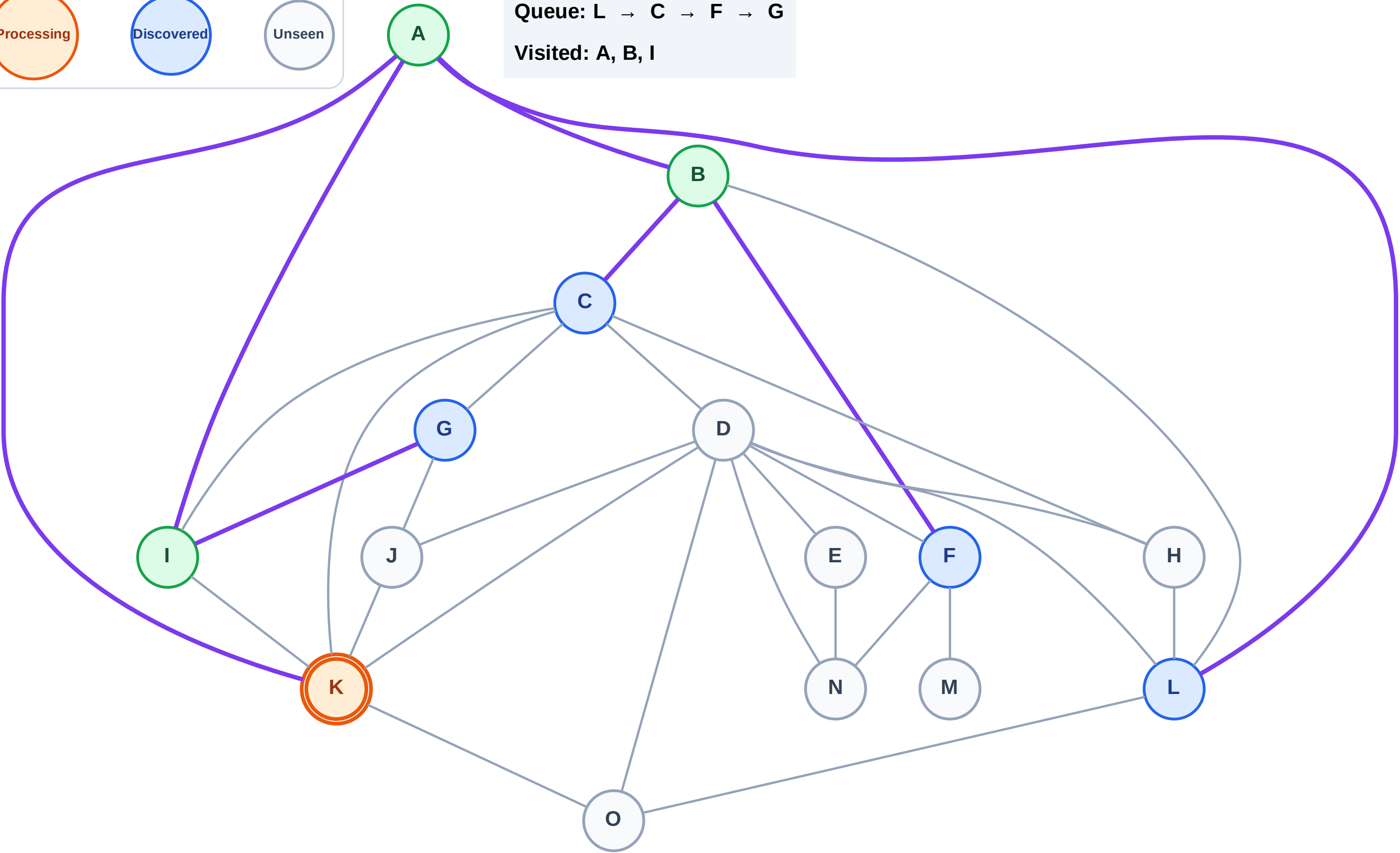
Processing

Discovered

Unseen

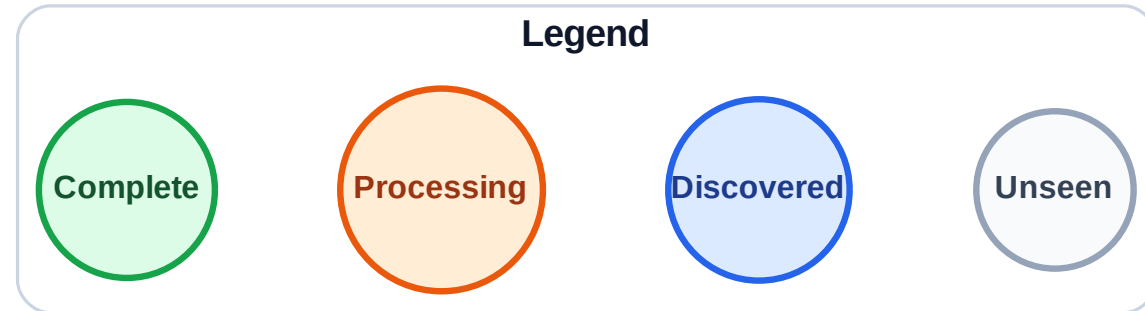
Queue: L → C → F → G

Visited: A, B, I

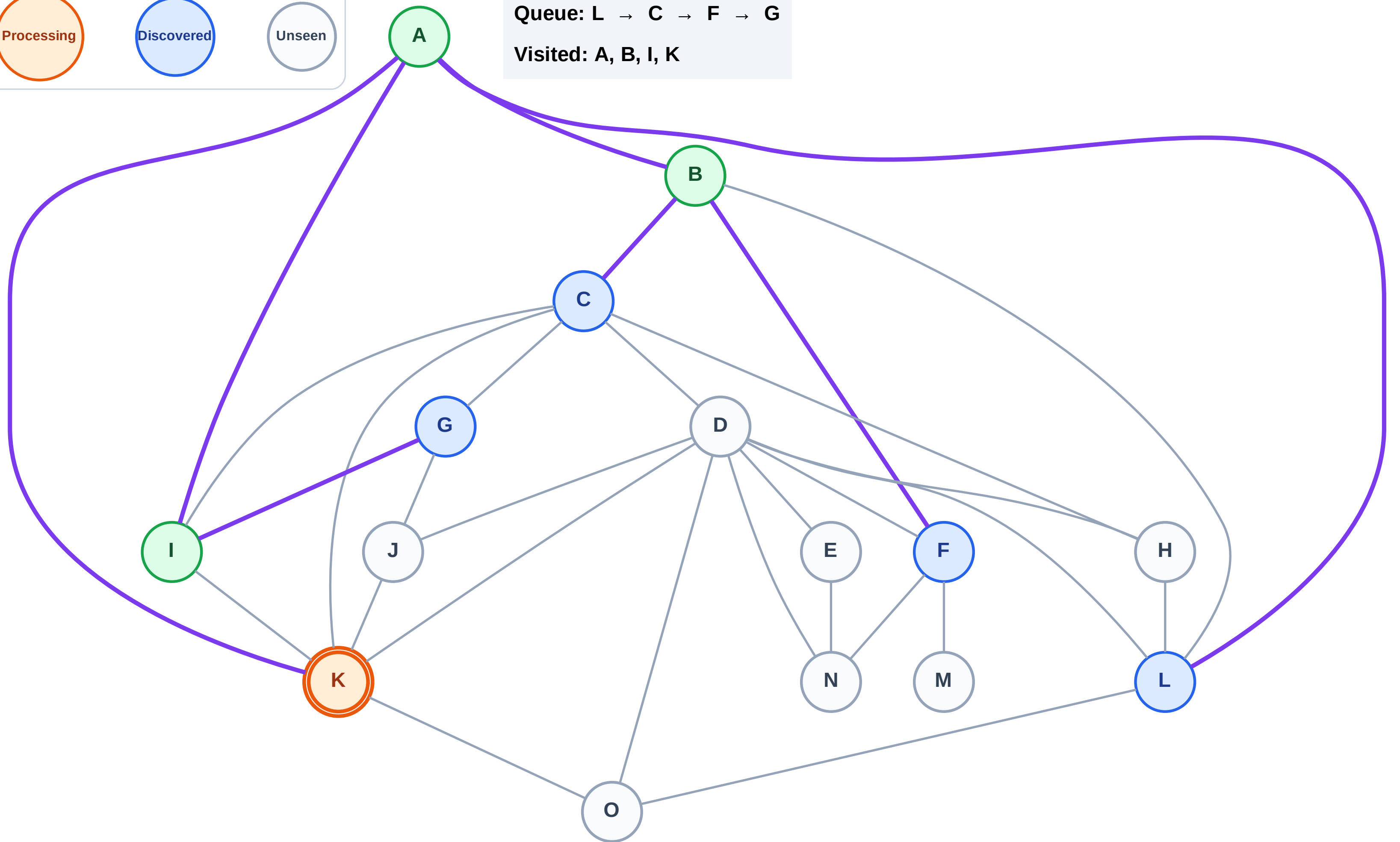


BFS Traversal

Step 9: TARGET NODE K REACHED! Path (1 hops): A → K

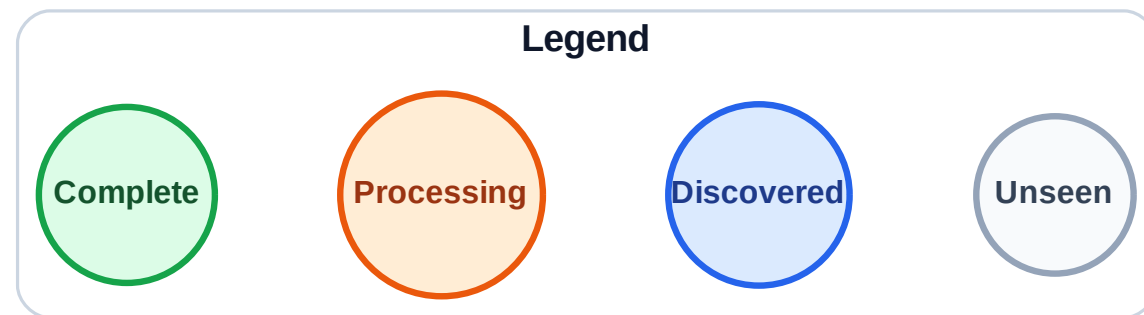


Queue: L → C → F → G
Visited: A, B, I, K



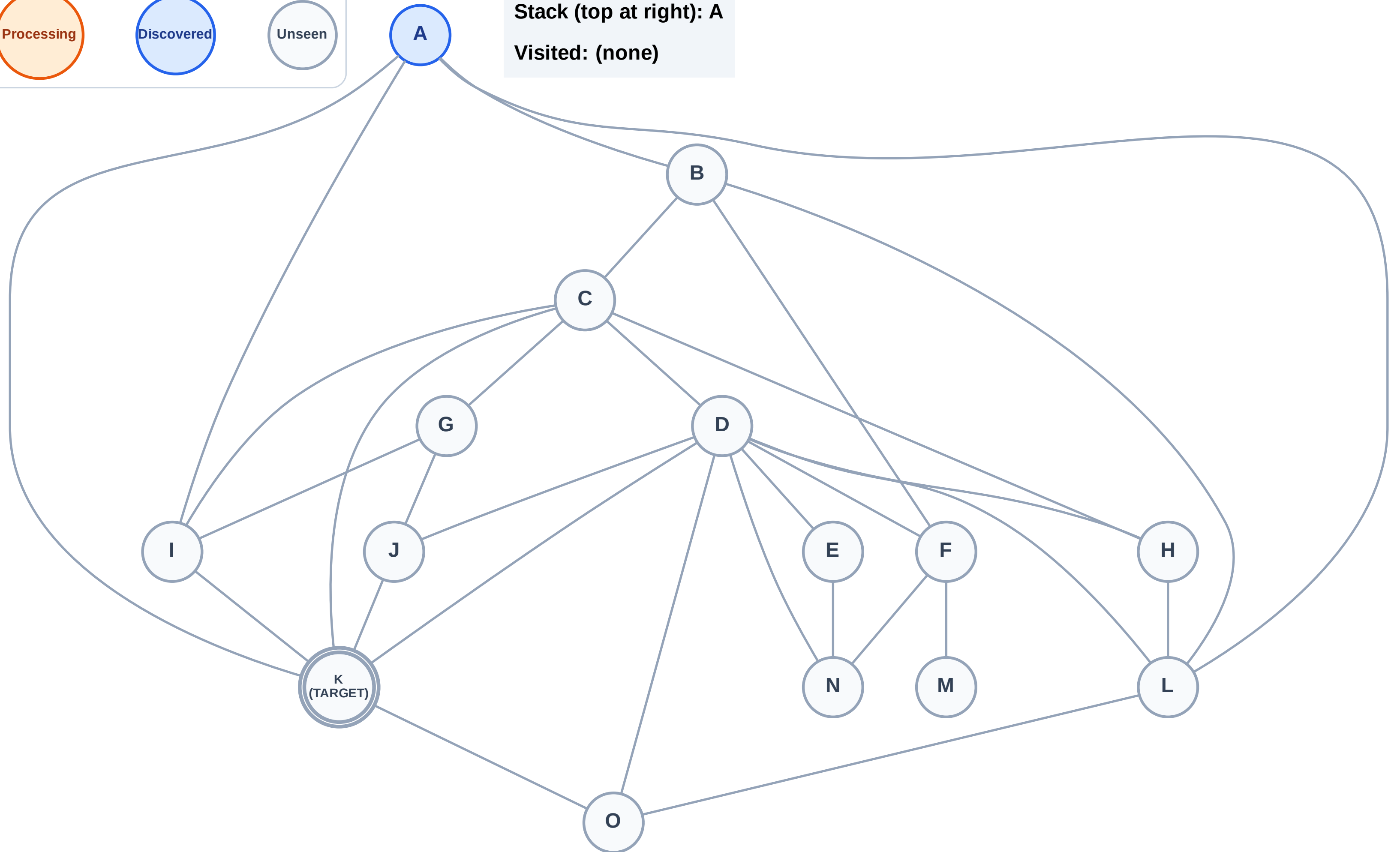
DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

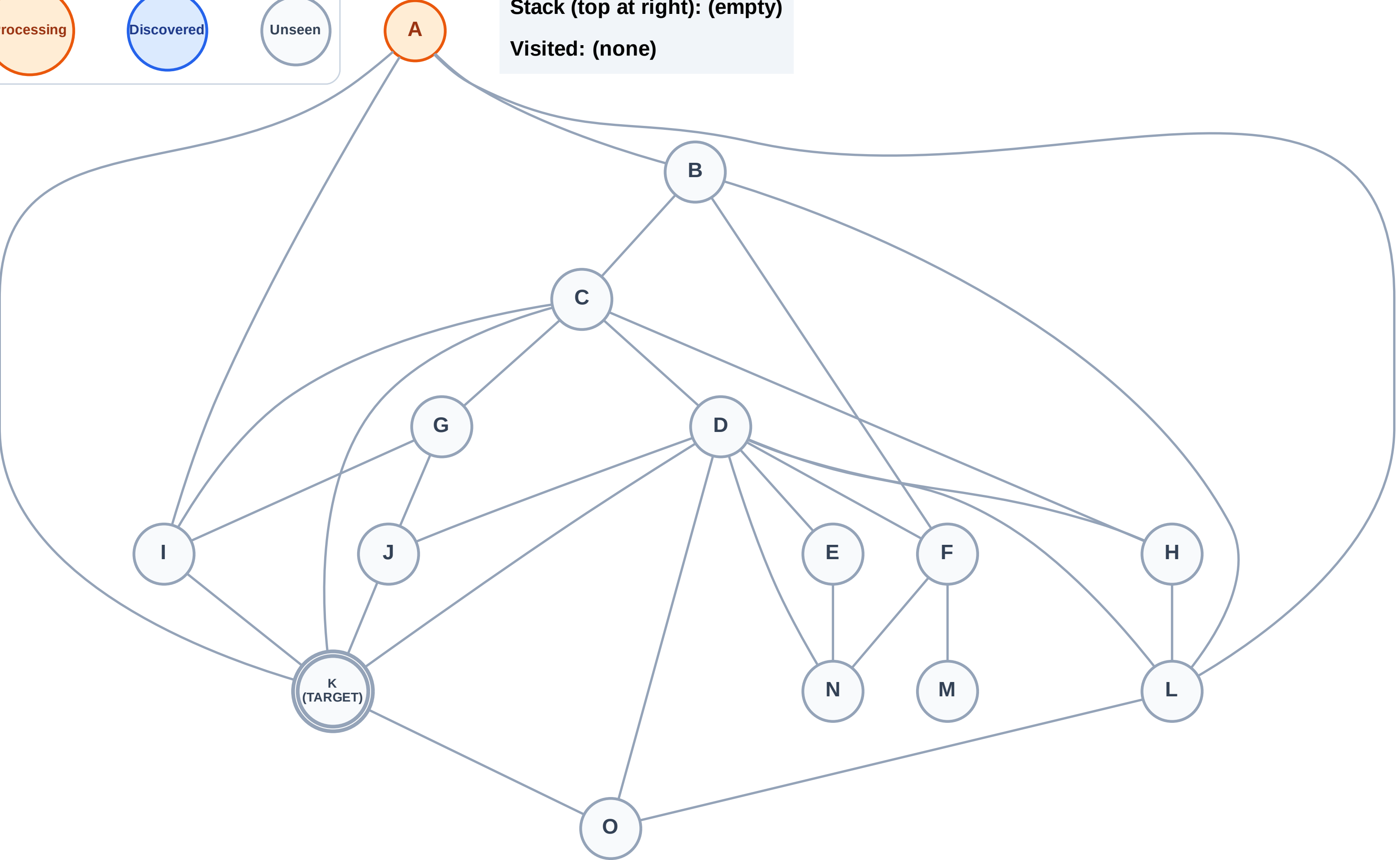
Processing

Discovered

Unseen

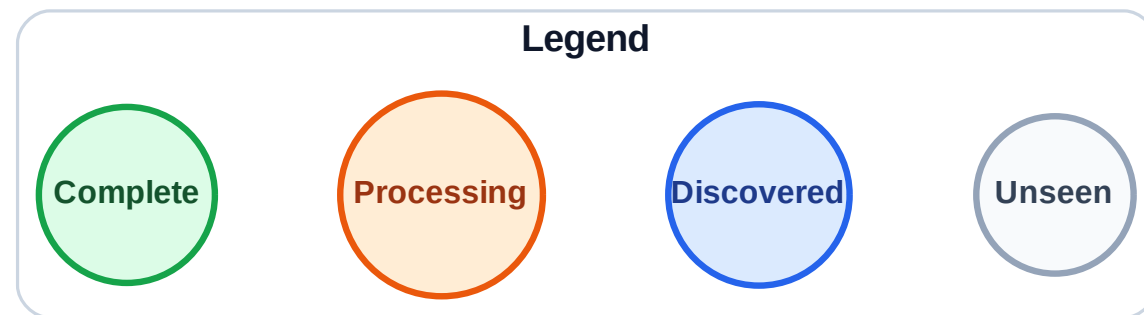
Stack (top at right): (empty)

Visited: (none)



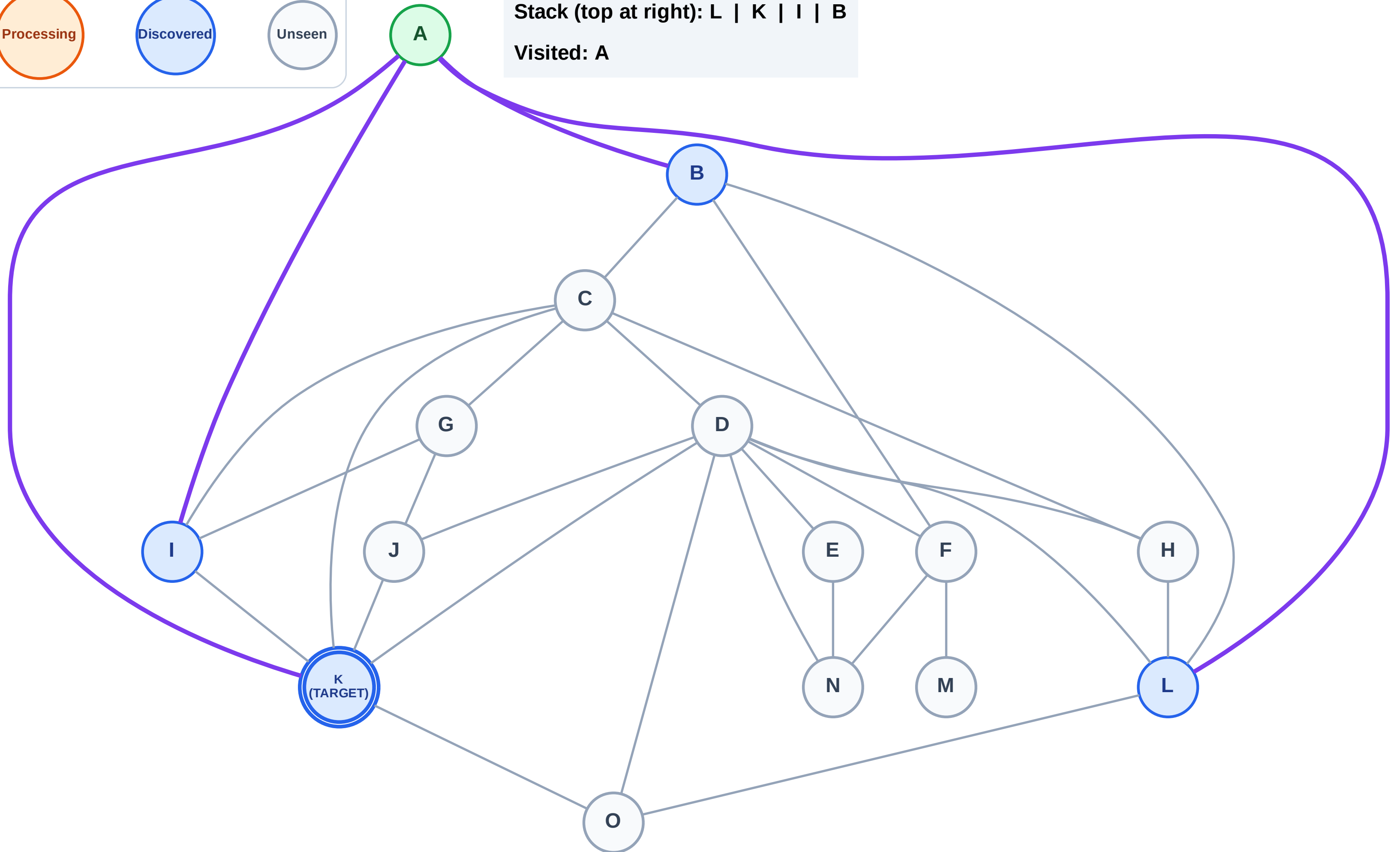
DFS Traversal

Step 3: Discover L, K, I, B from A



Stack (top at right): L | K | I | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

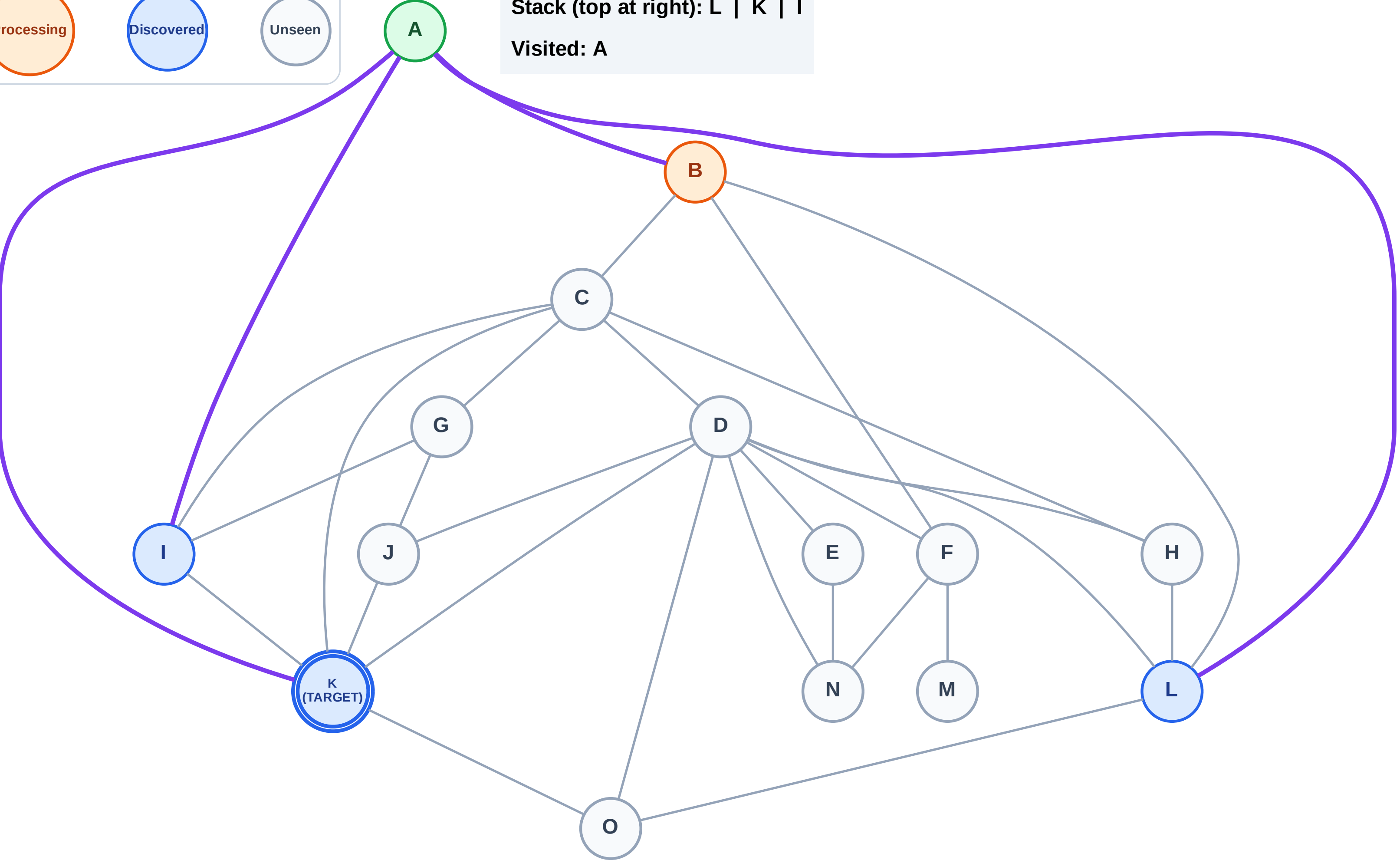
Complete

Processing

Discovered

Unseen

Stack (top at right): L | K | I
Visited: A



DFS Traversal

Step 5: Discover F, C from B

Stack (top at right): L | K | I | F | C
Visited: A, B

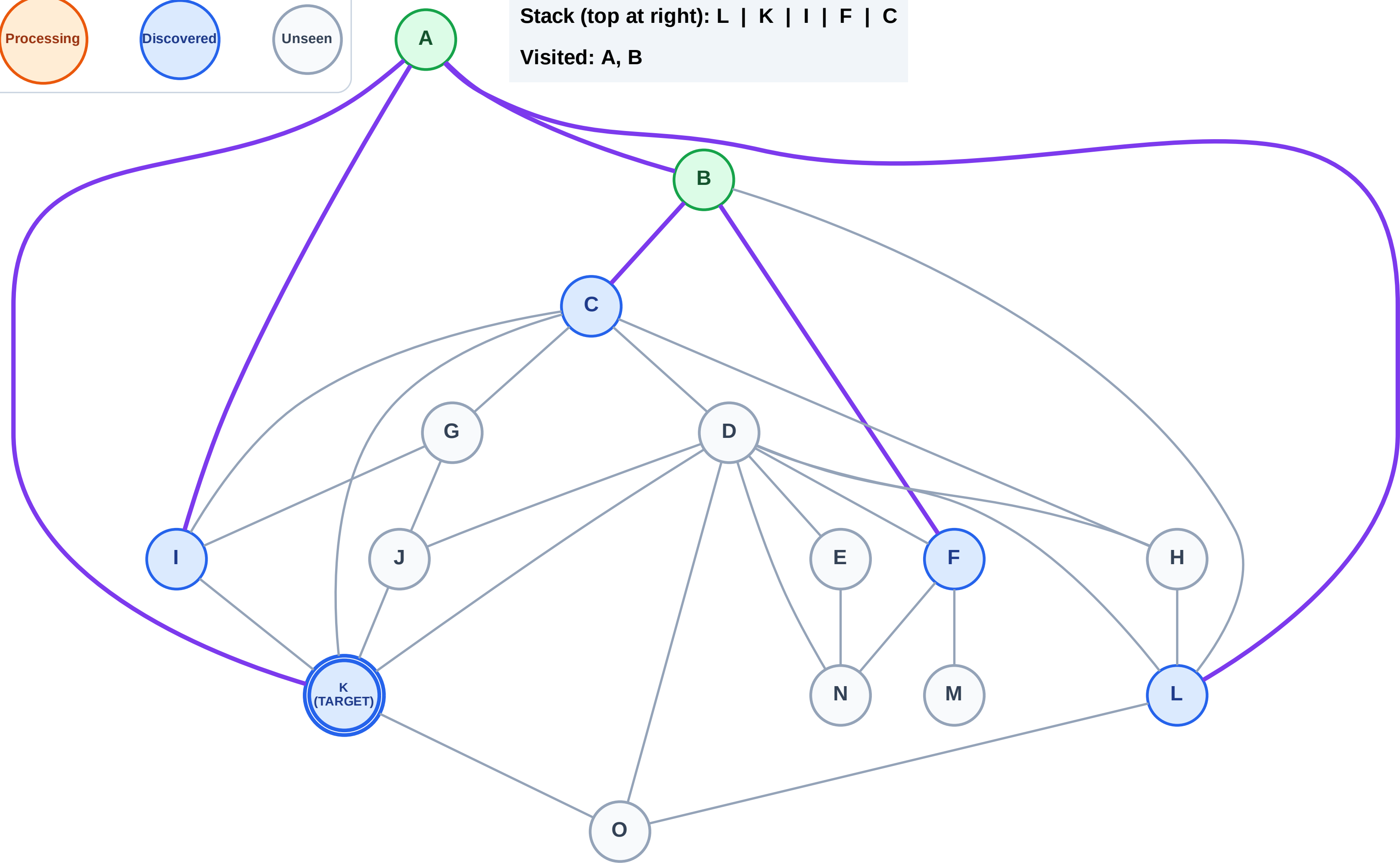
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 6: Process node C

Legend

Complete

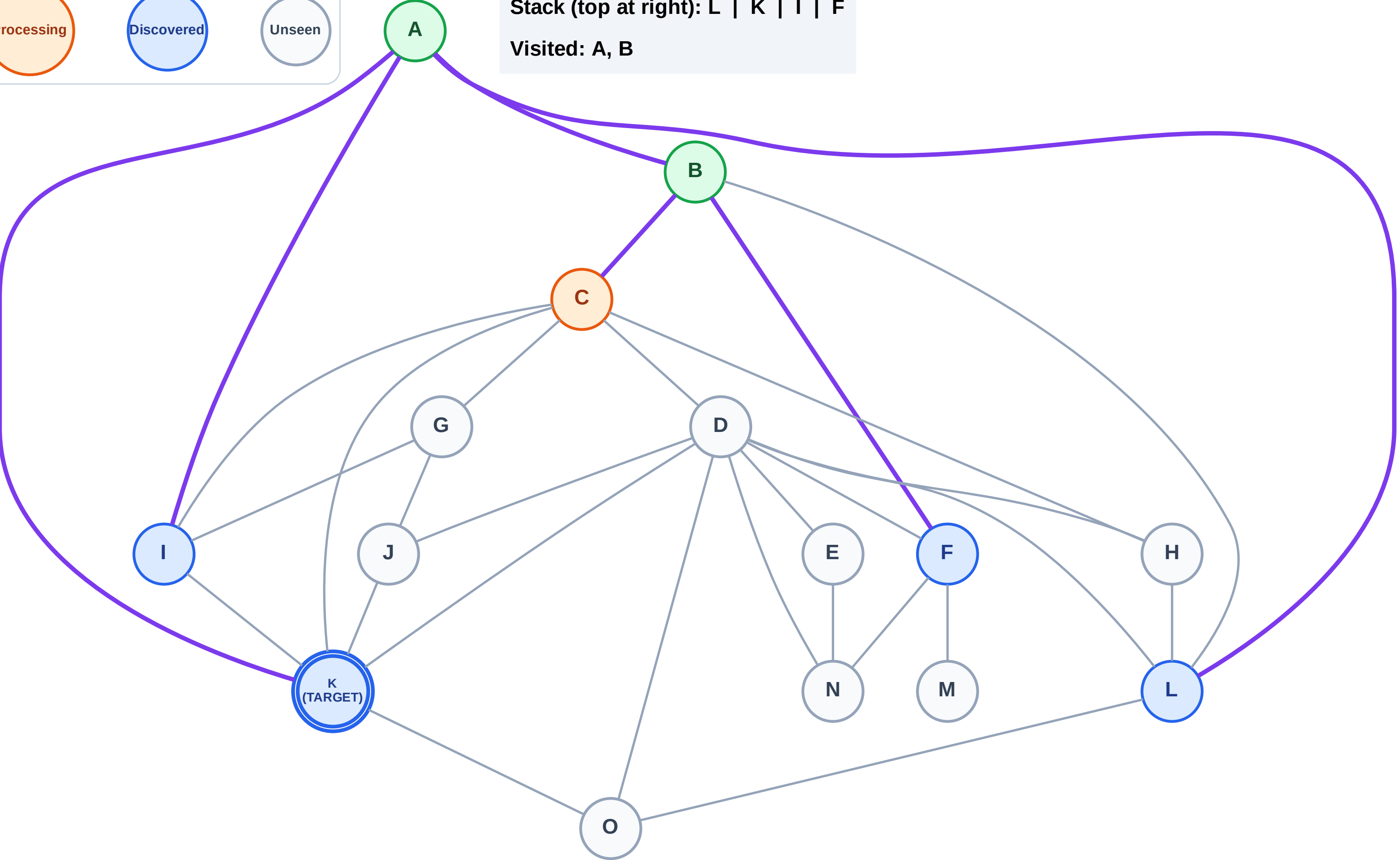
Processing

Discovered

Unseen

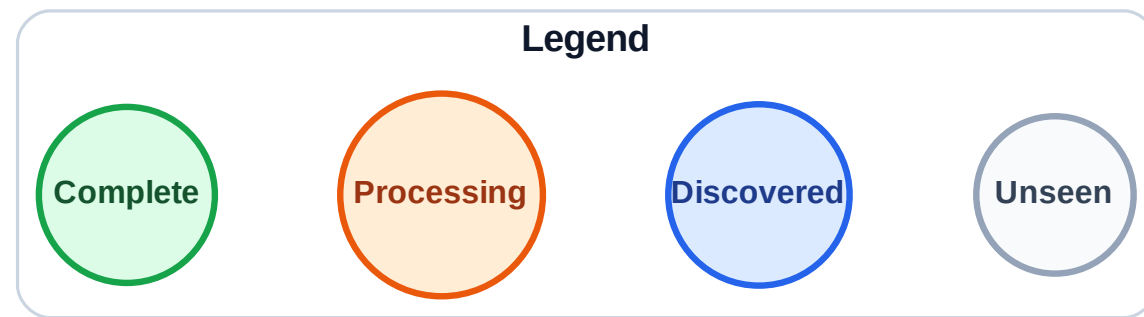
Stack (top at right): L | K | I | F

Visited: A, B

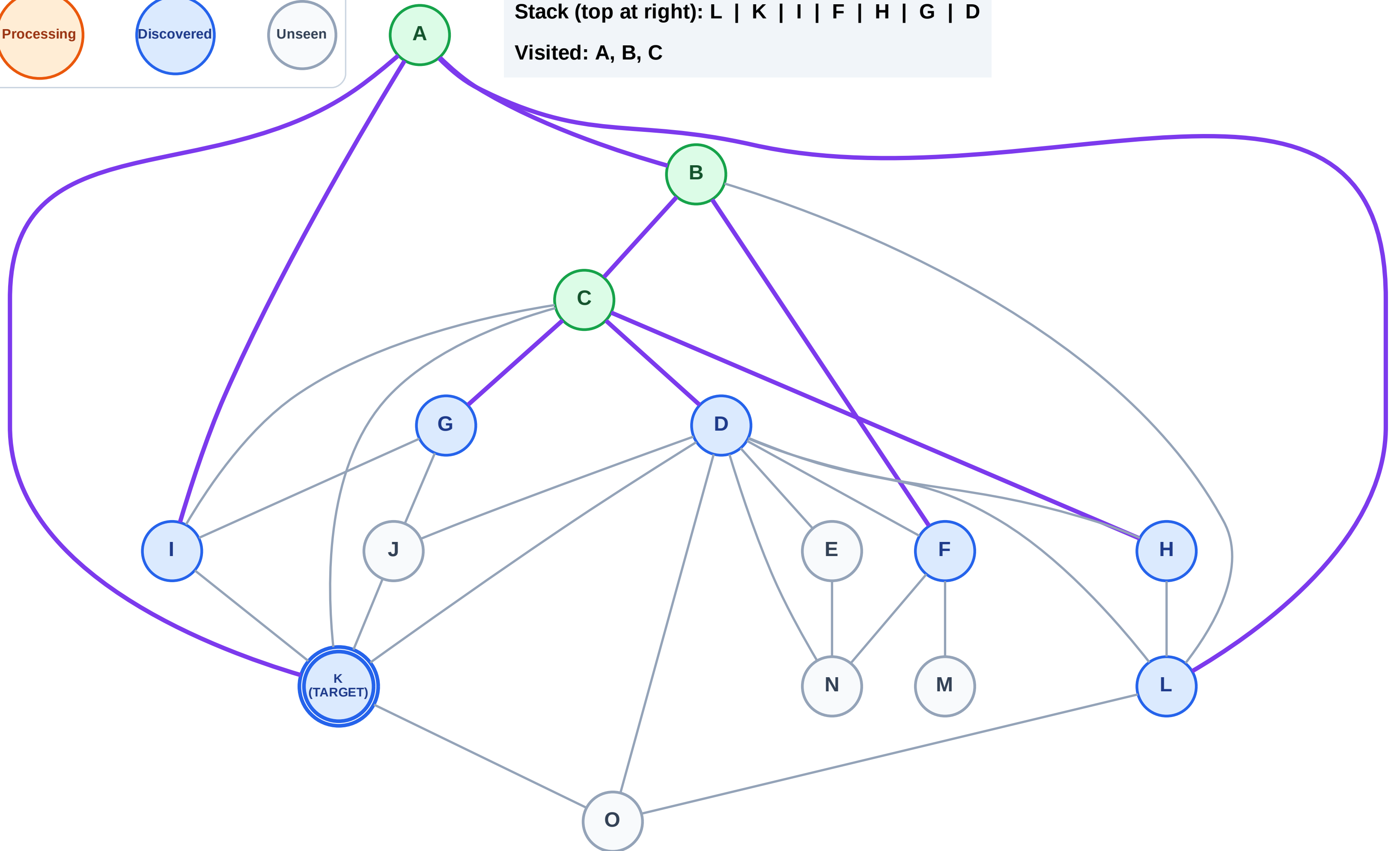


DFS Traversal

Step 7: Discover H, G, D from C



Stack (top at right): L | K | I | F | H | G | D
Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

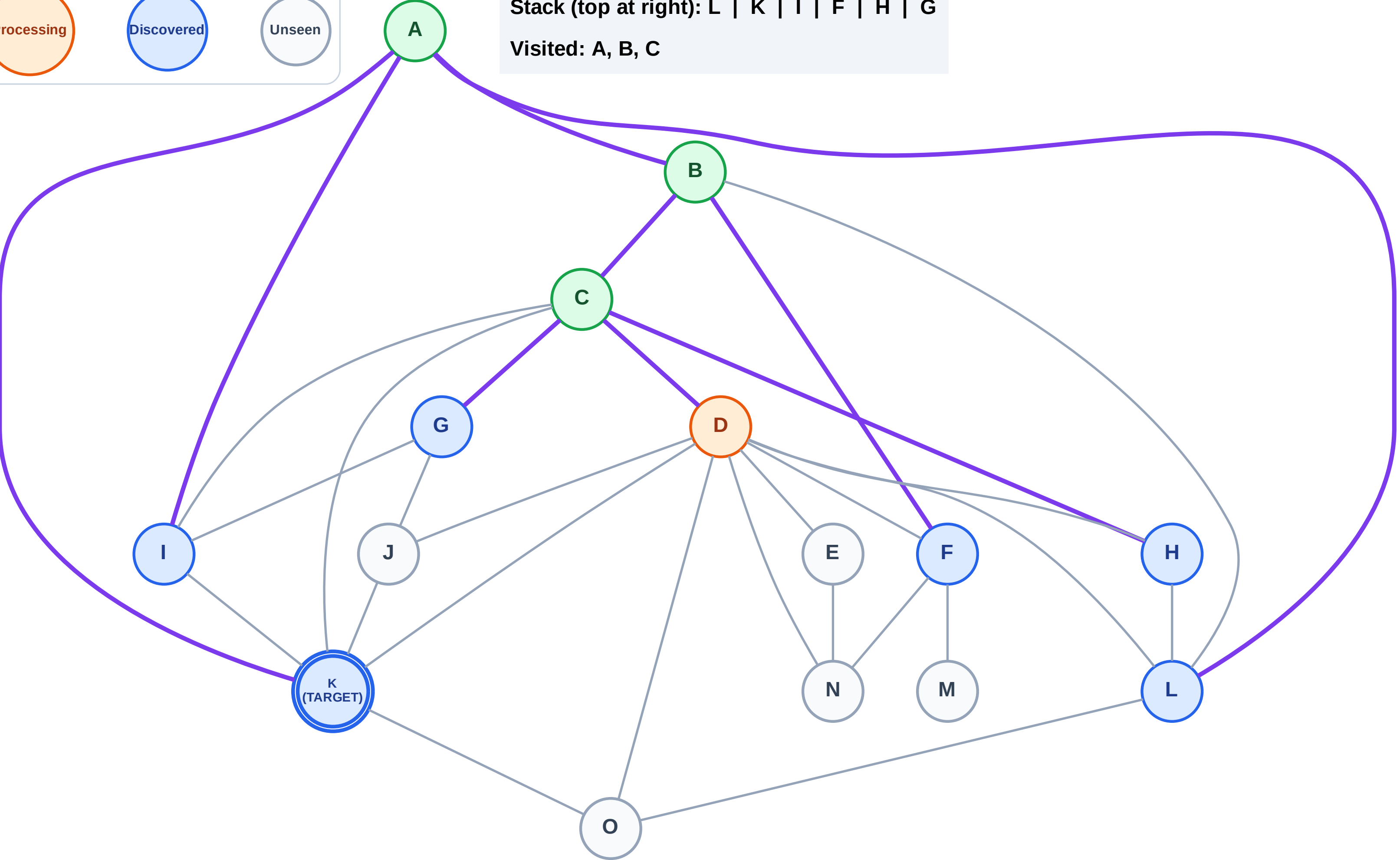
Processing

Discovered

Unseen

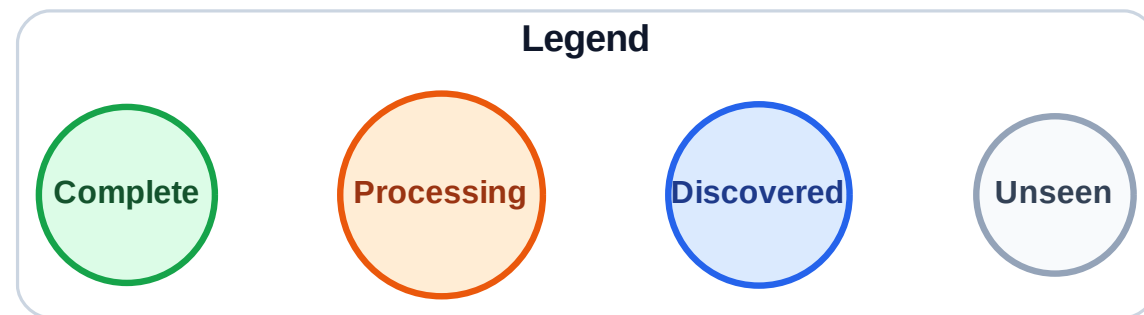
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C



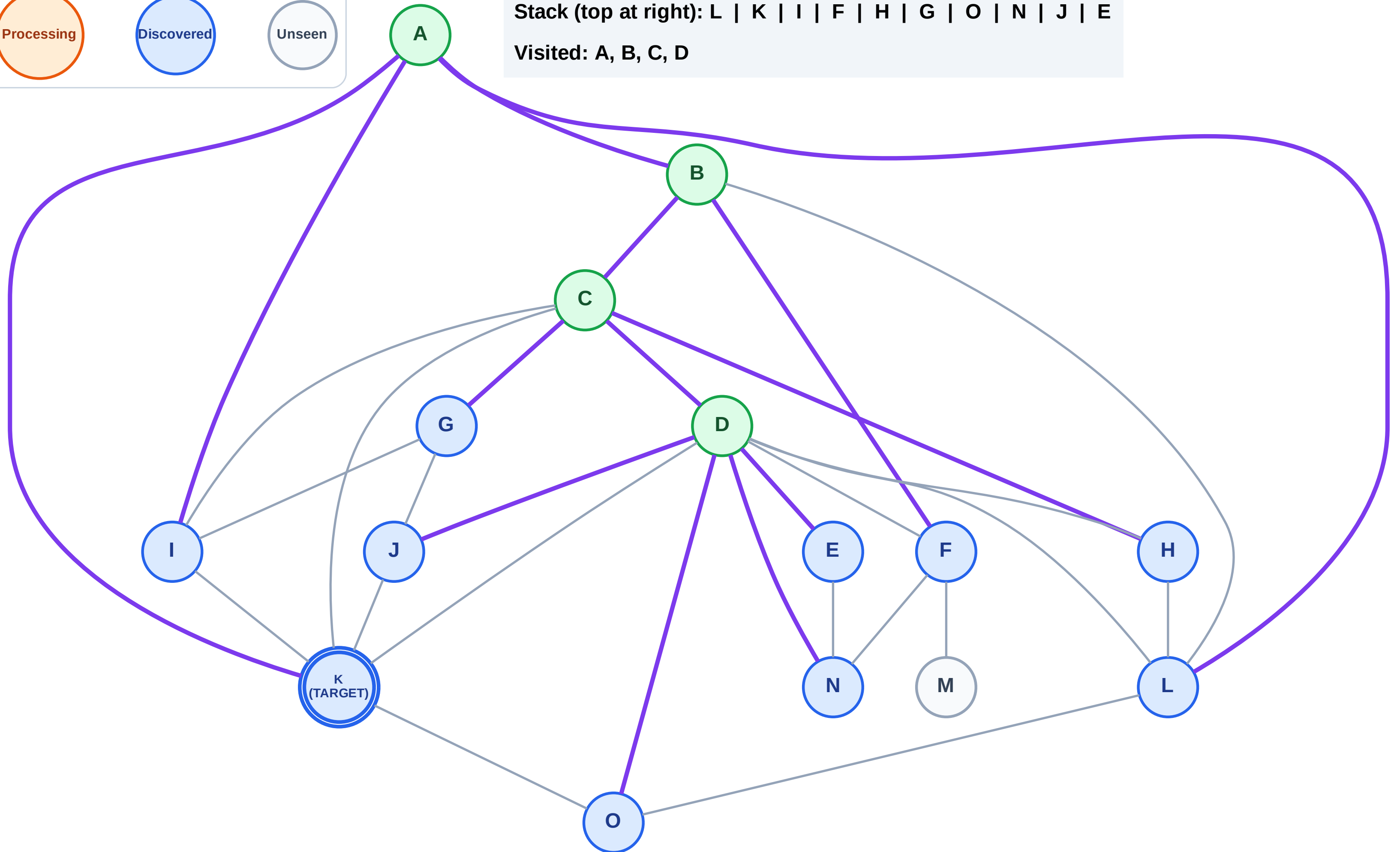
DFS Traversal

Step 9: Discover O, N, J, E from D

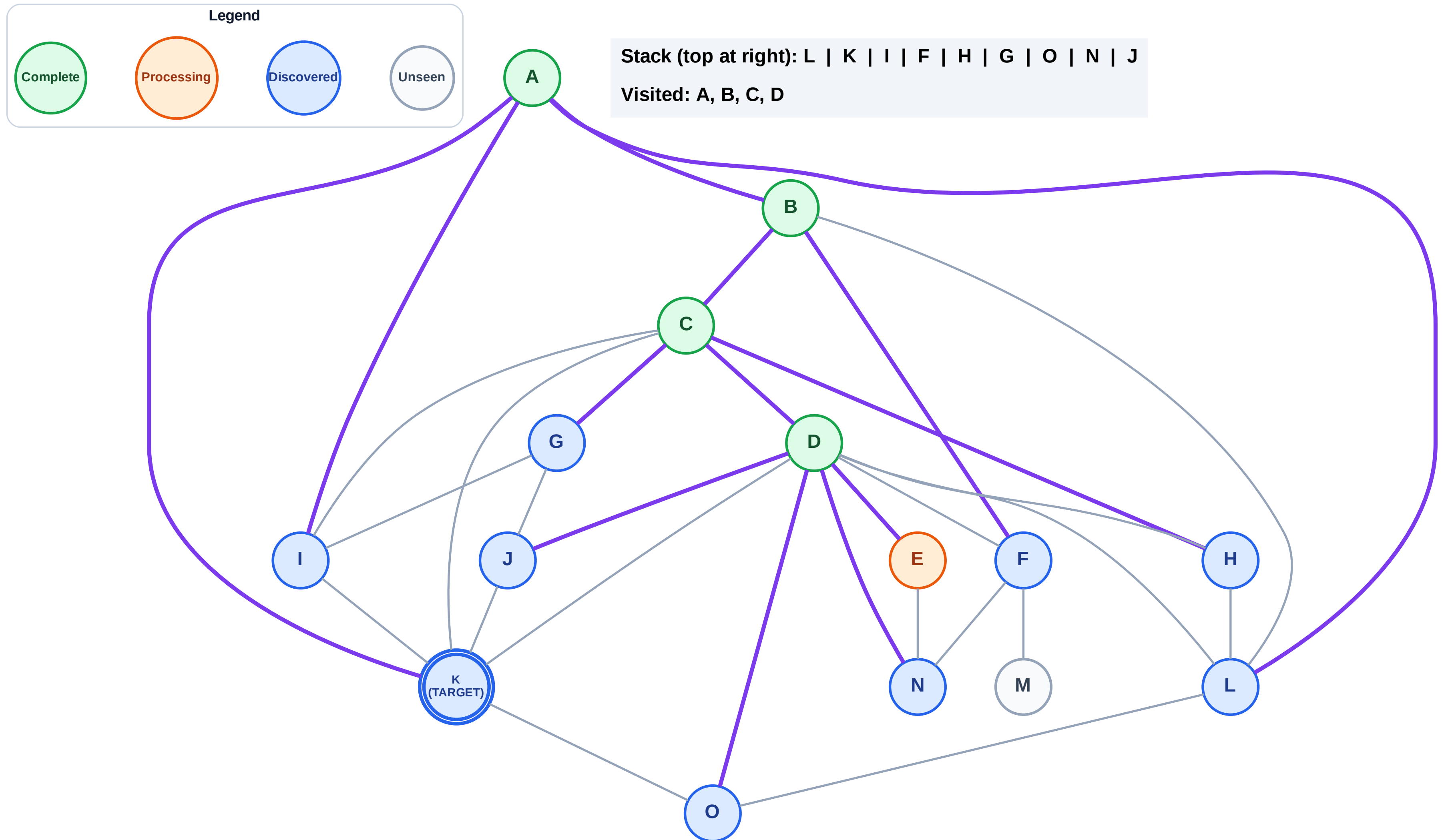


Stack (top at right): L | K | I | F | H | G | O | N | J | E

Visited: A, B, C, D



Step 10: Process node E

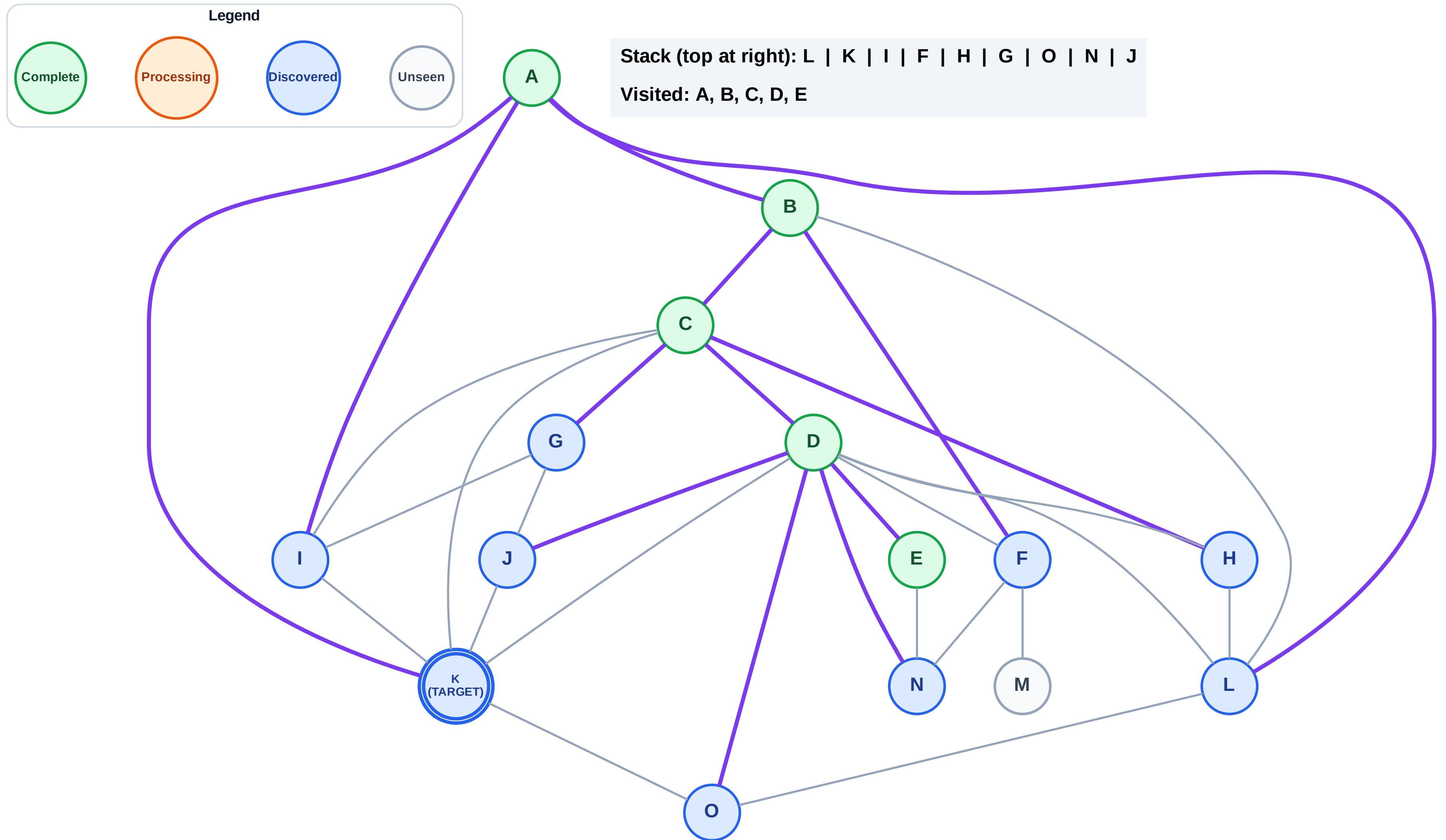


DFS Traversal

Step 11: No new neighbors from E

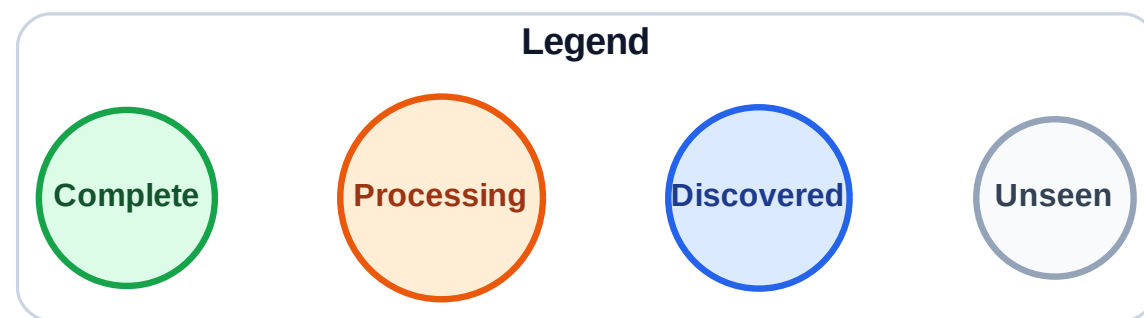
DFS Traversal

Step 11: No new neighbors from E



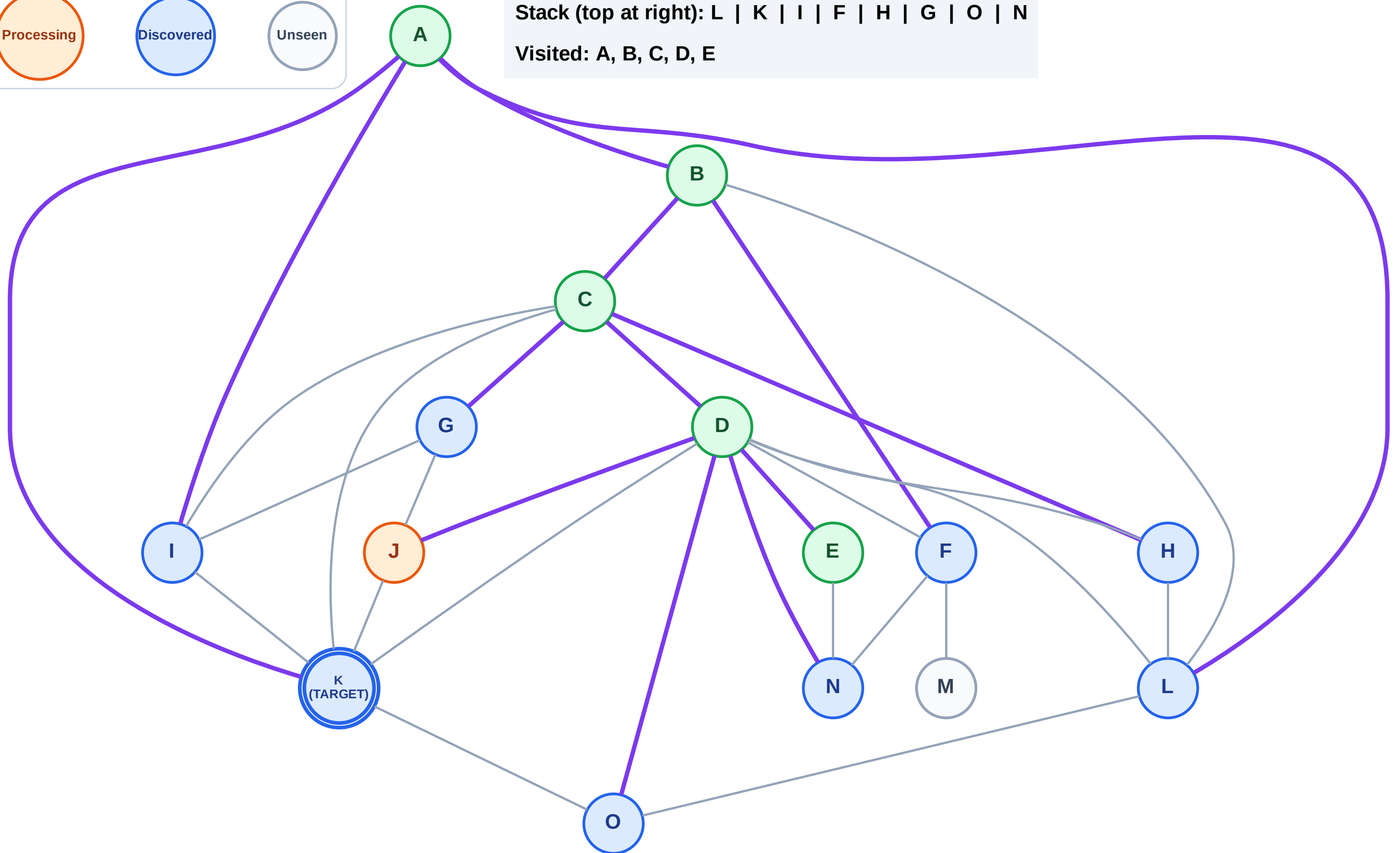
DFS Traversal

Step 12: Process node J



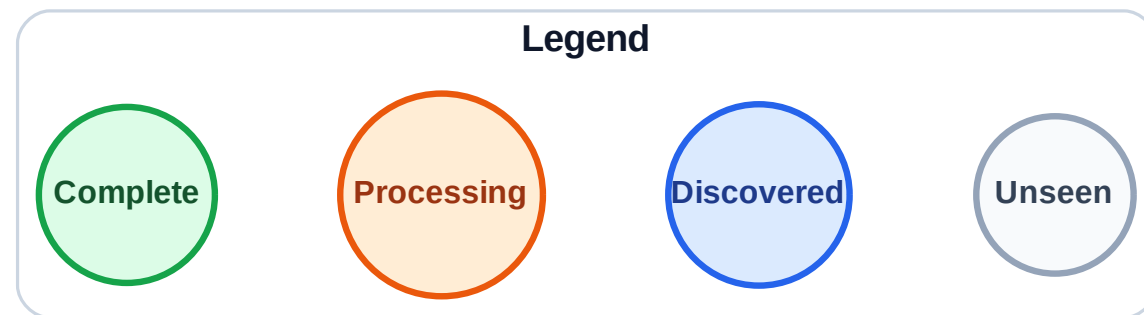
Stack (top at right): L | K | I | F | H | G | O | N

Visited: A, B, C, D, E



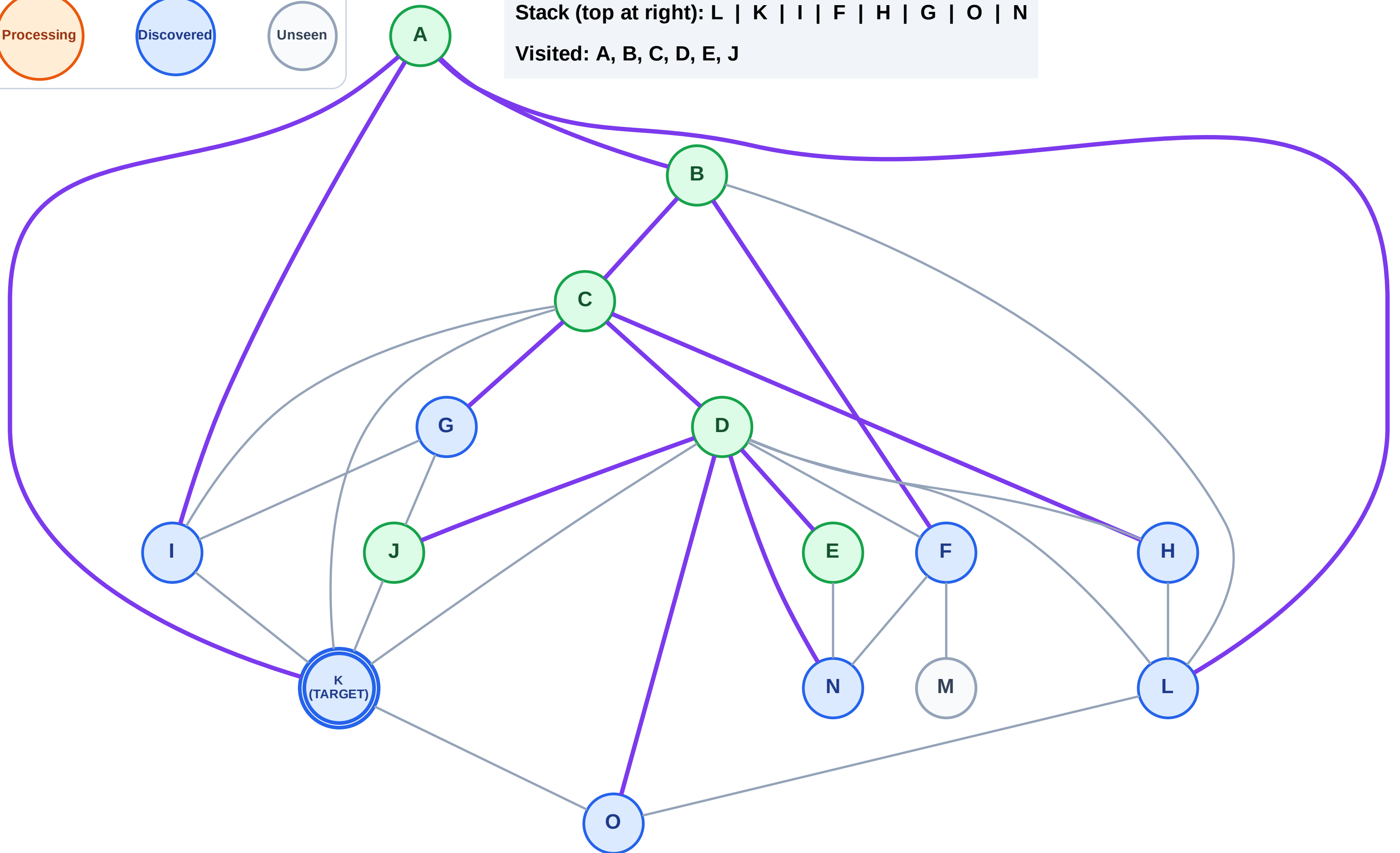
DFS Traversal

Step 13: No new neighbors from J



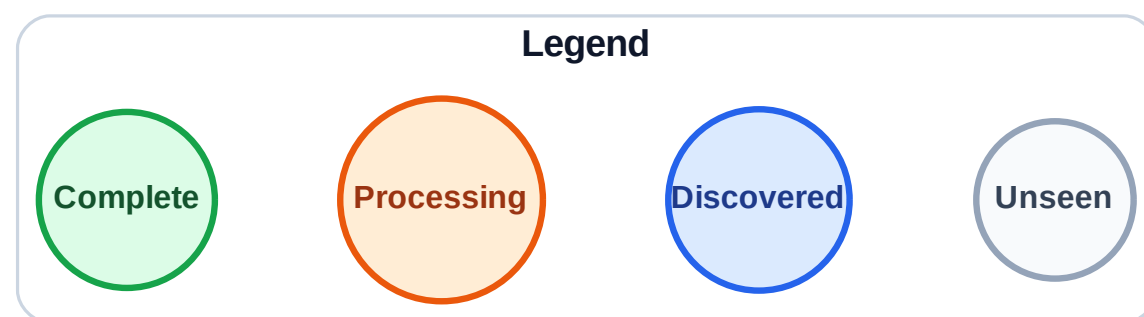
Stack (top at right): L | K | I | F | H | G | O | N

Visited: A, B, C, D, E, J



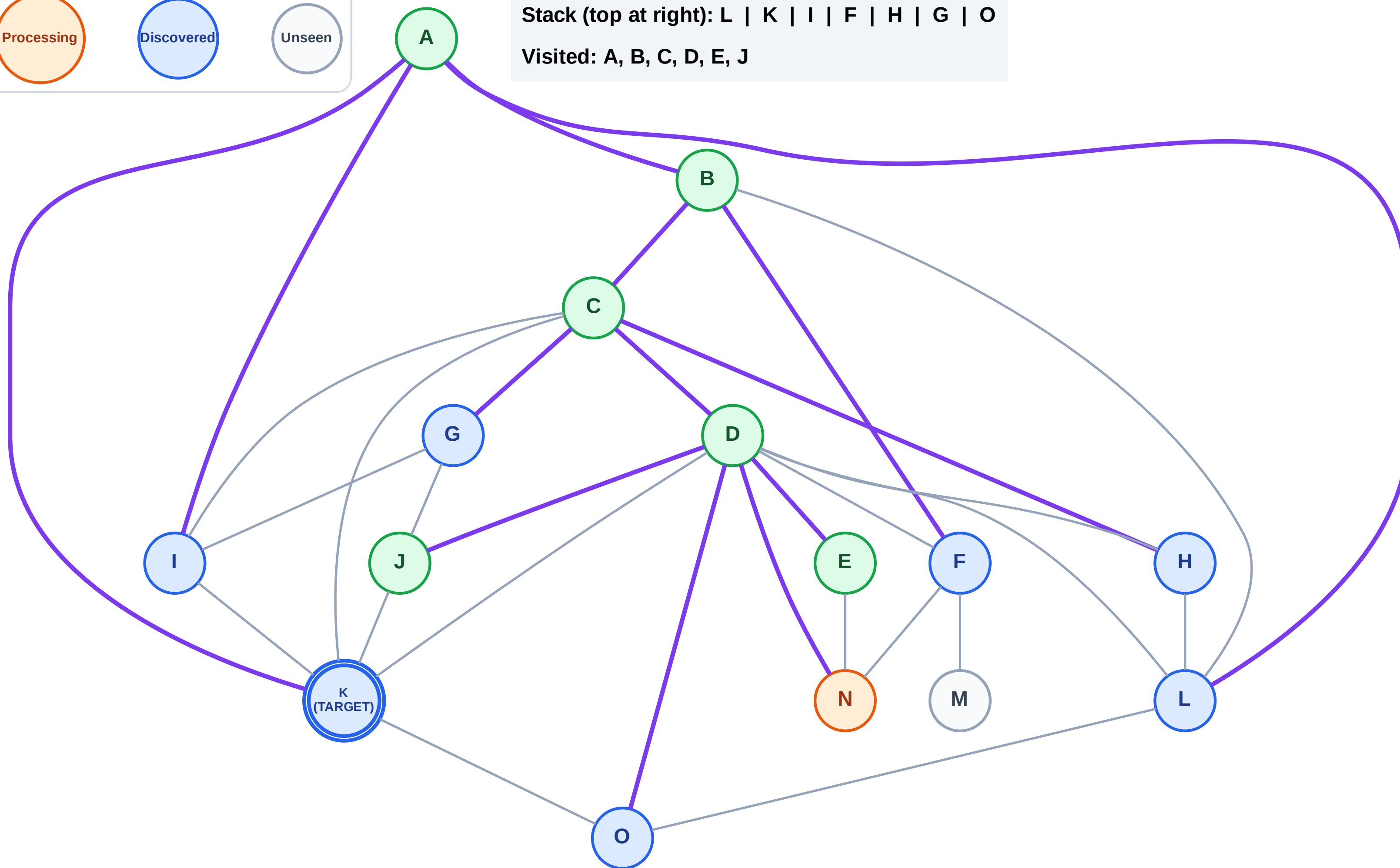
DFS Traversal

Step 14: Process node N



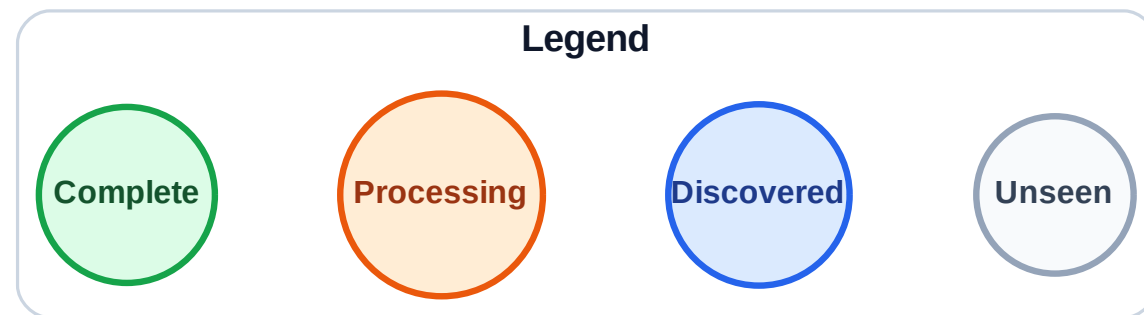
Stack (top at right): L | K | I | F | H | G | O

Visited: A, B, C, D, E, J



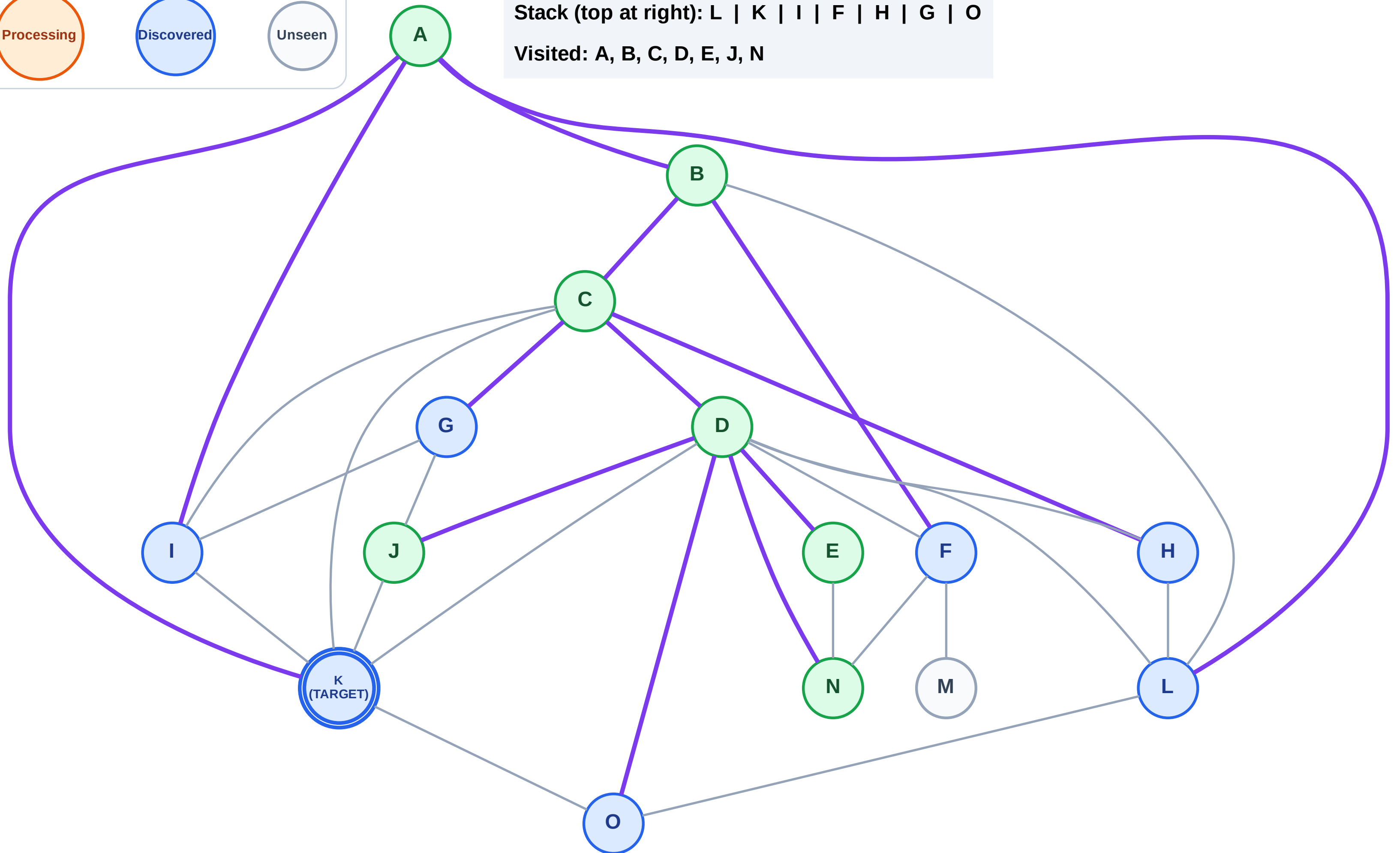
DFS Traversal

Step 15: No new neighbors from N



Stack (top at right): L | K | I | F | H | G | O

Visited: A, B, C, D, E, J, N



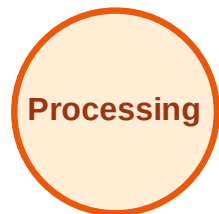
DFS Traversal

Step 16: Process node O

Legend



Complete



Processing



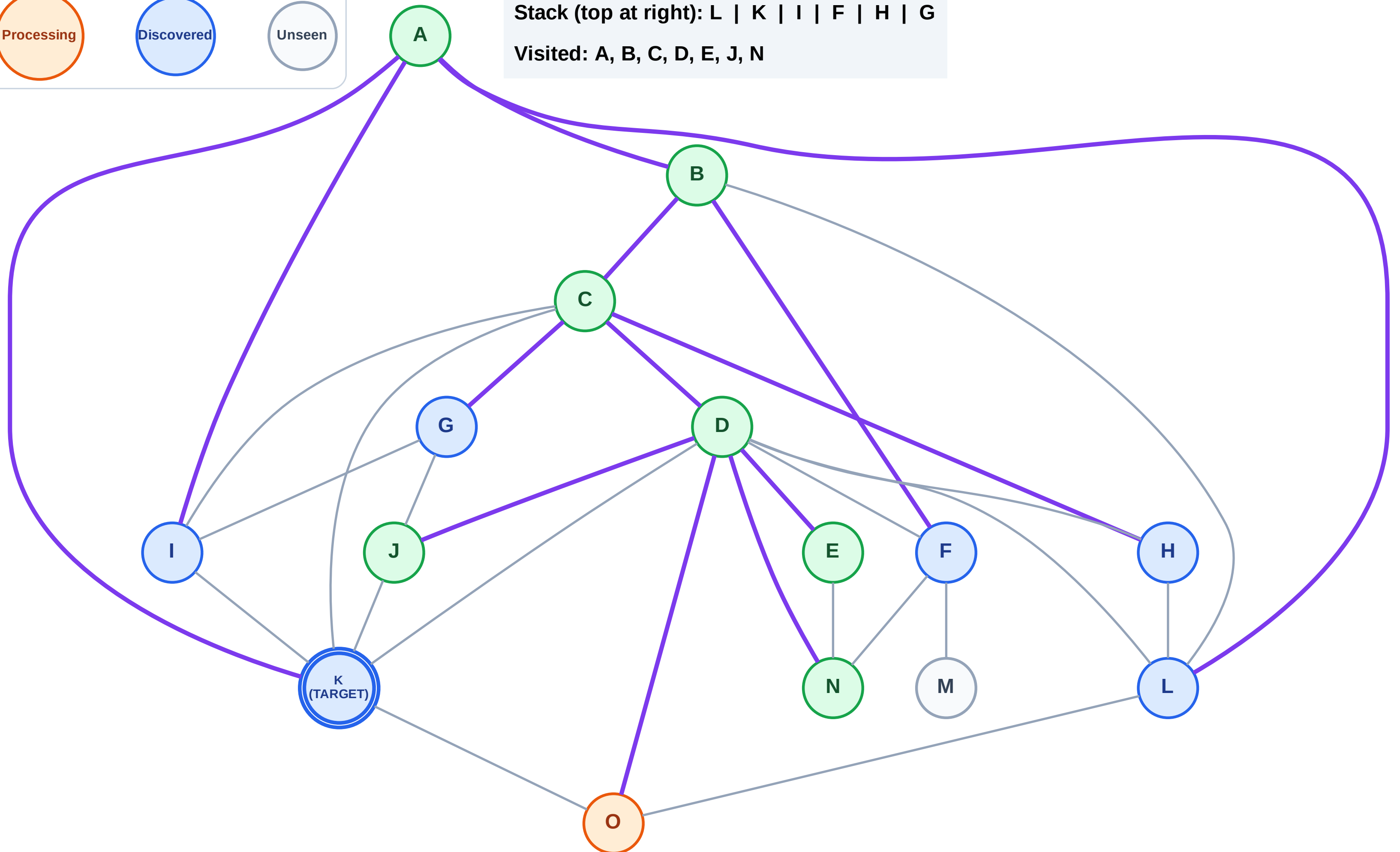
Discovered



Unseen

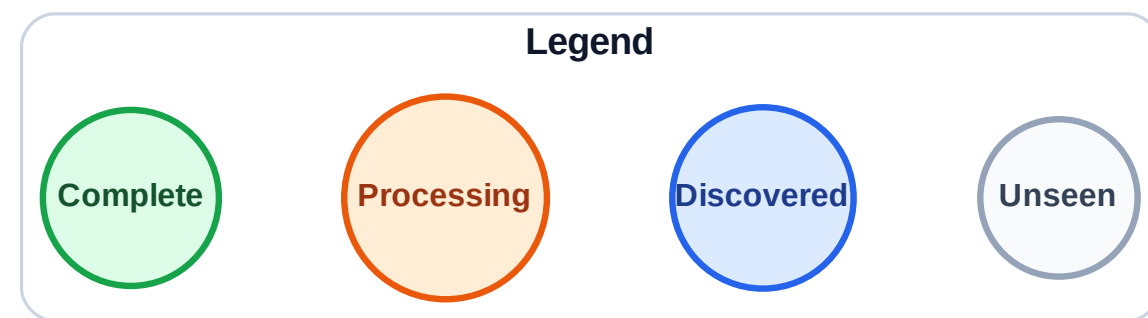
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C, D, E, J, N



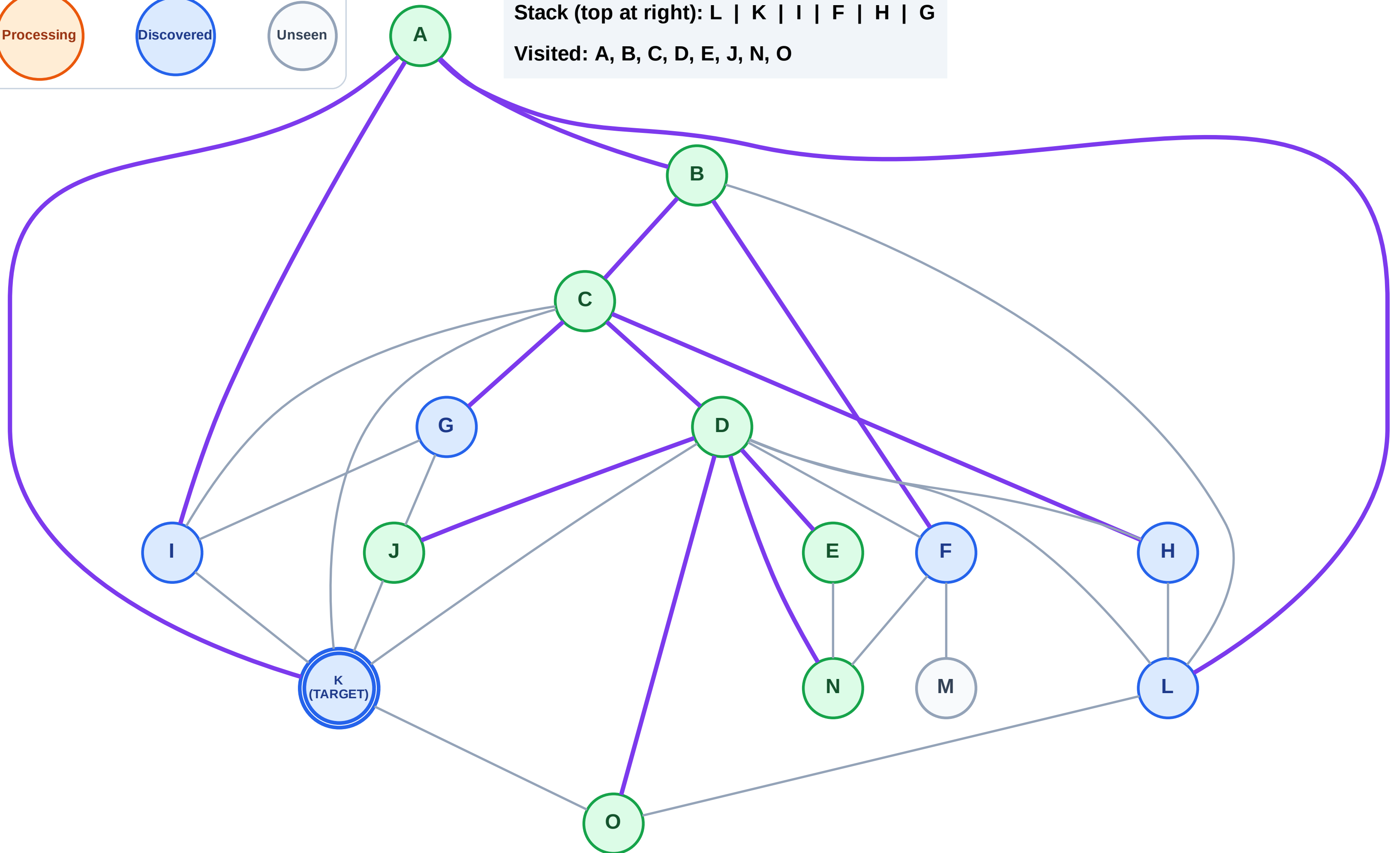
DFS Traversal

Step 17: No new neighbors from O



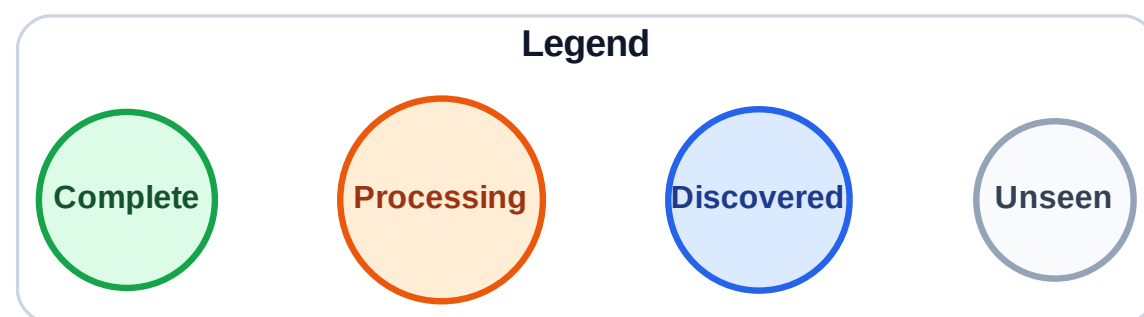
Stack (top at right): L | K | I | F | H | G

Visited: A, B, C, D, E, J, N, O



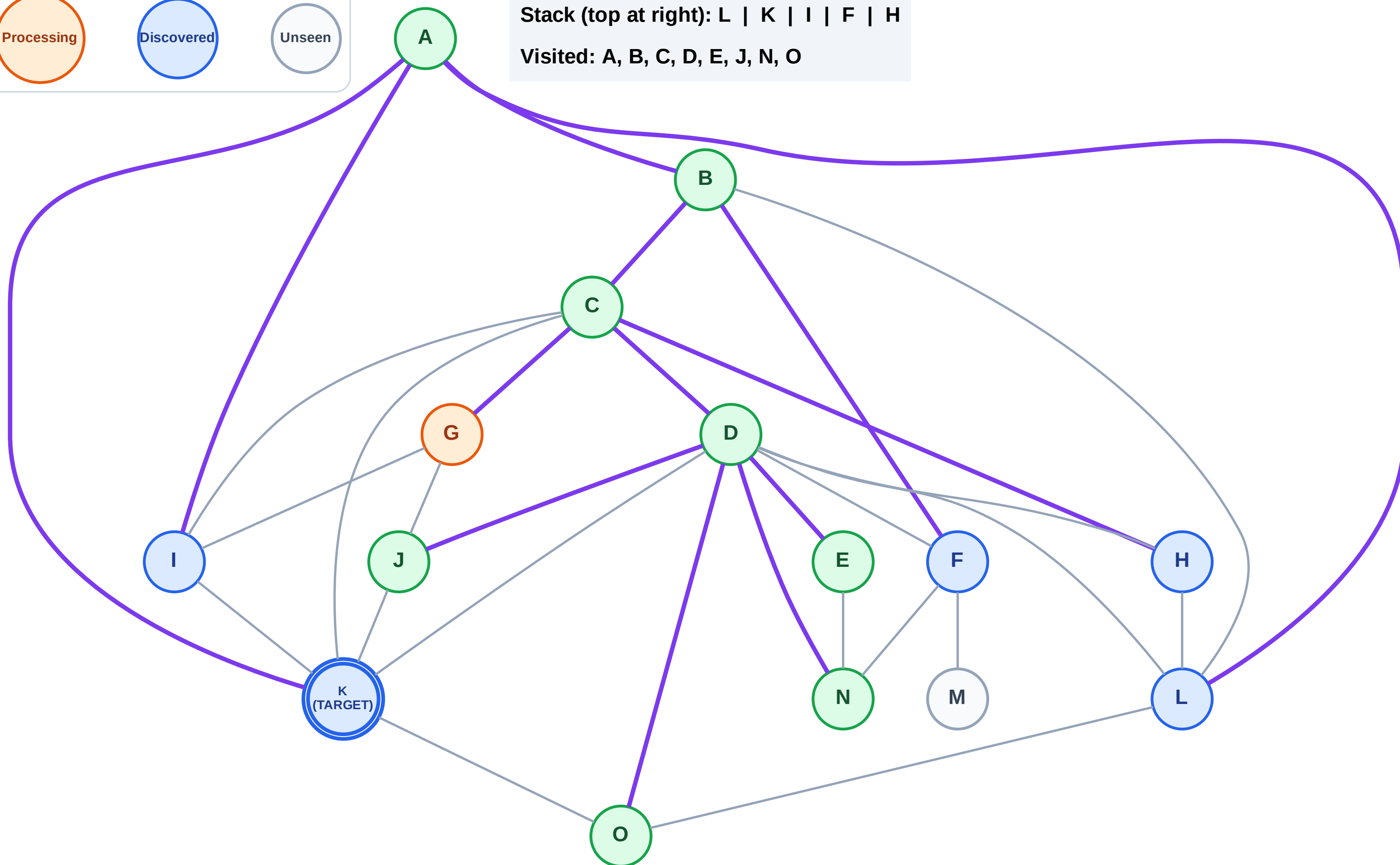
DFS Traversal

Step 18: Process node G



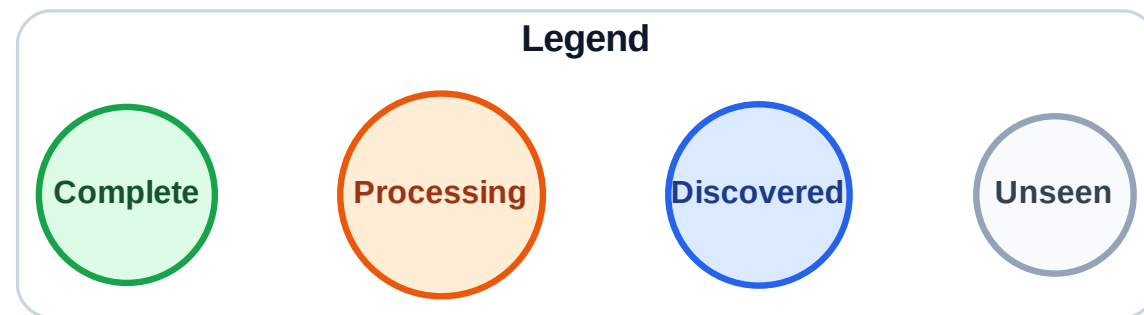
Stack (top at right): L | K | I | F | H

Visited: A, B, C, D, E, J, N, O



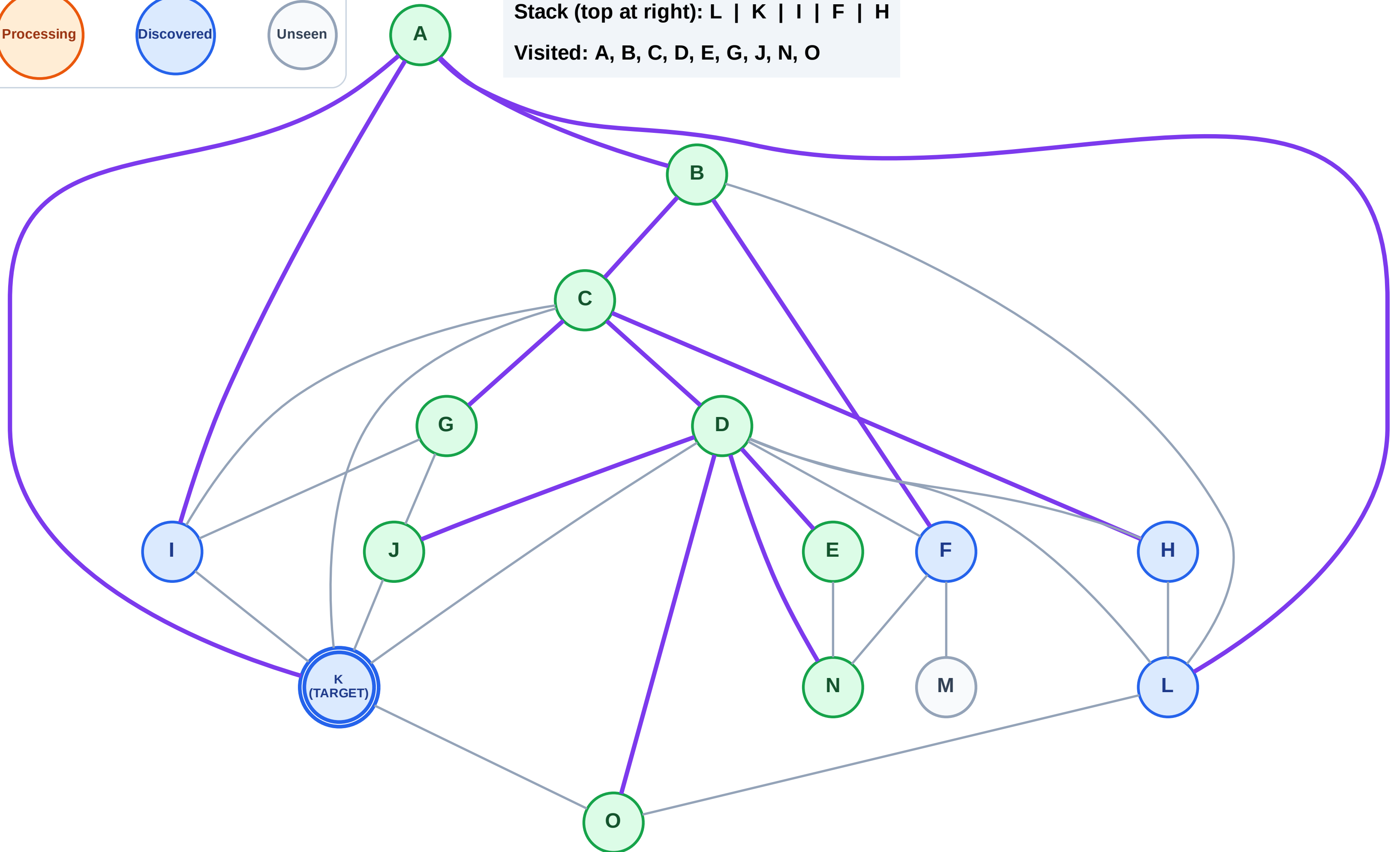
DFS Traversal

Step 19: No new neighbors from G



Stack (top at right): L | K | I | F | H

Visited: A, B, C, D, E, G, J, N, O



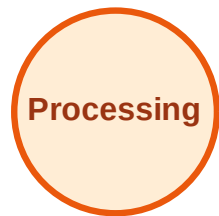
DFS Traversal

Step 20: Process node H

Legend



Complete



Processing



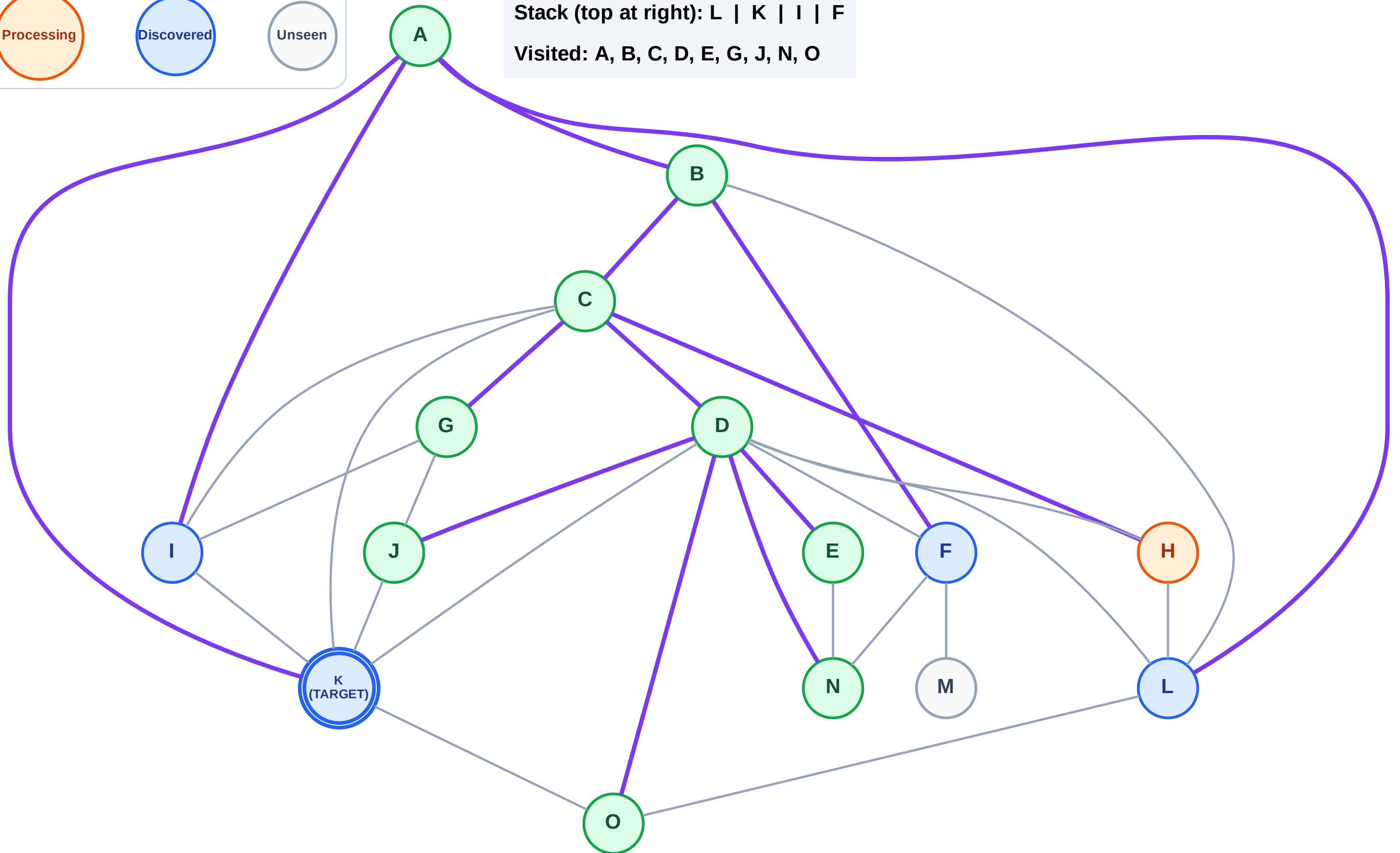
Discovered



Unseen

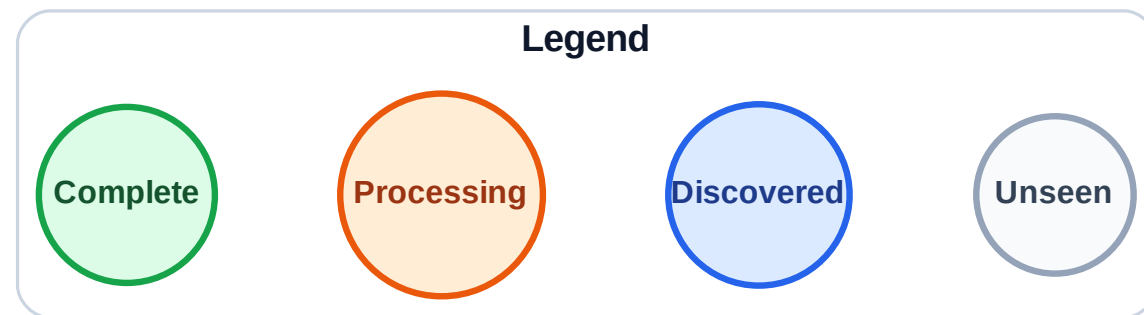
Stack (top at right): L | K | I | F

Visited: A, B, C, D, E, G, J, N, O



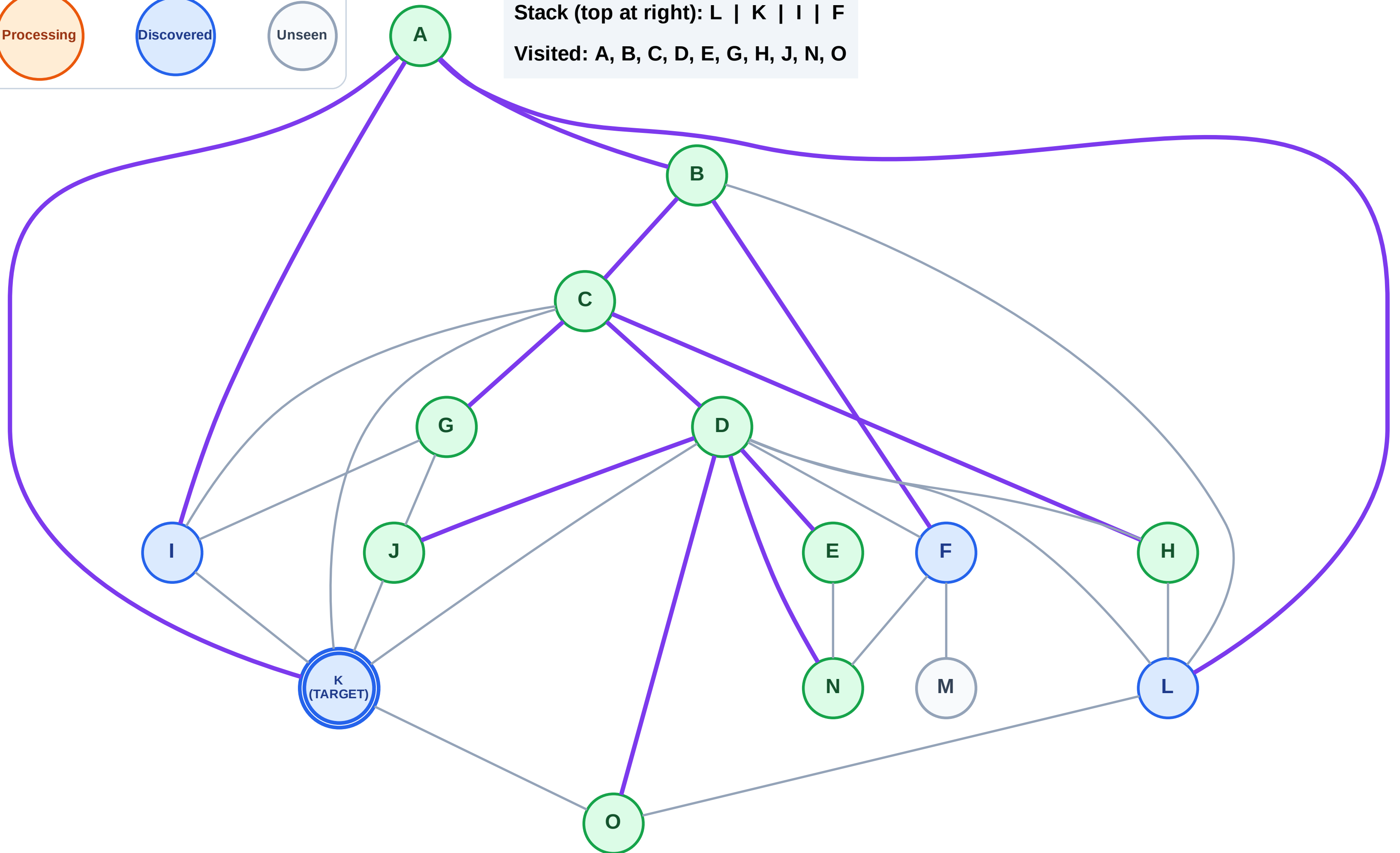
DFS Traversal

Step 21: No new neighbors from H



Stack (top at right): L | K | I | F

Visited: A, B, C, D, E, G, H, J, N, O



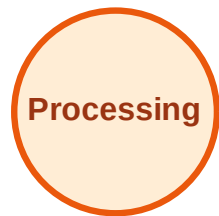
DFS Traversal

Step 22: Process node F

Legend



Complete



Processing



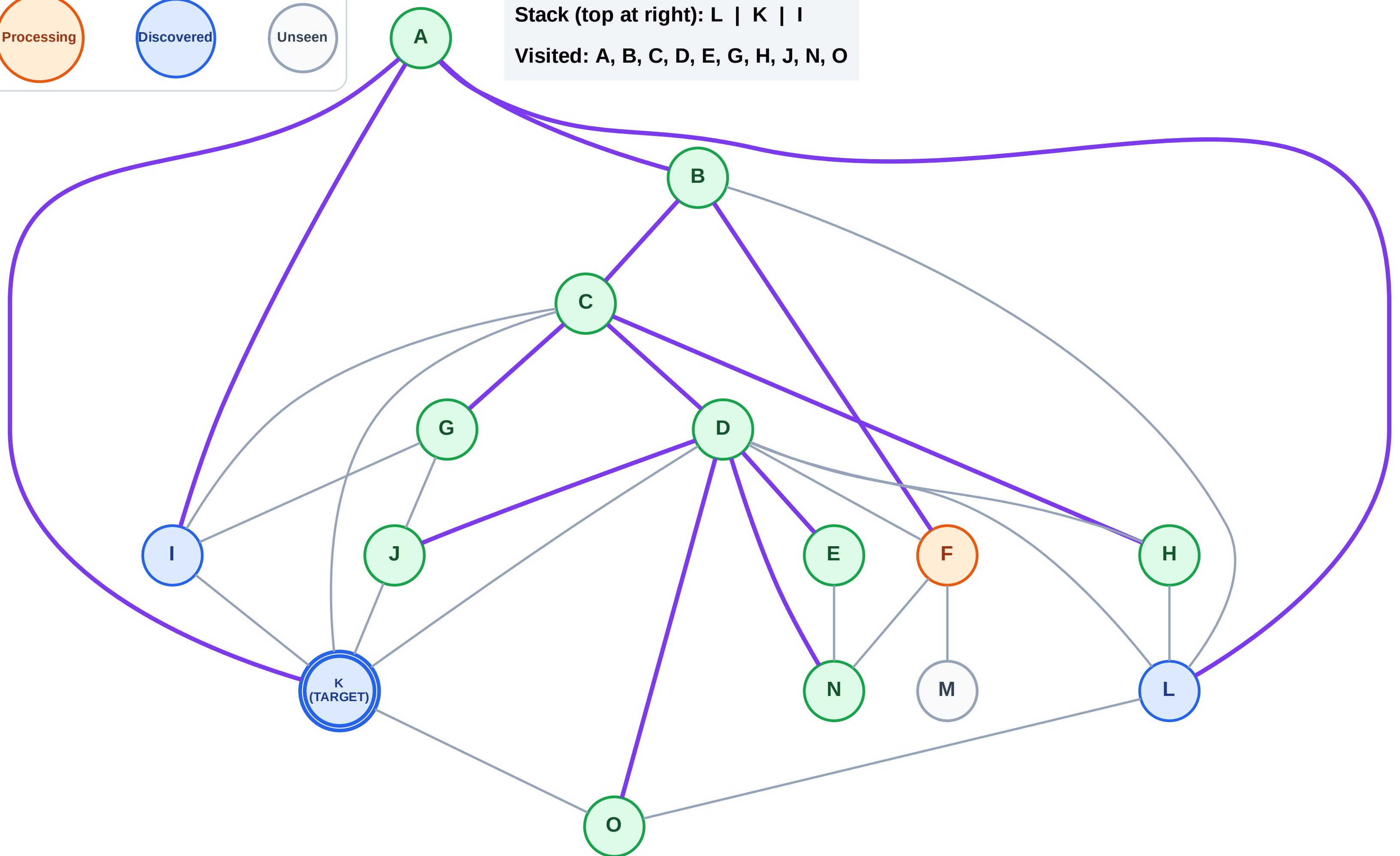
Discovered



Unseen

Stack (top at right): L | K | I

Visited: A, B, C, D, E, G, H, J, N, O



DFS Traversal

Step 23: Discover M from F

Stack (top at right): L | K | I | M
Visited: A, B, C, D, E, F, G, H, J, N, O

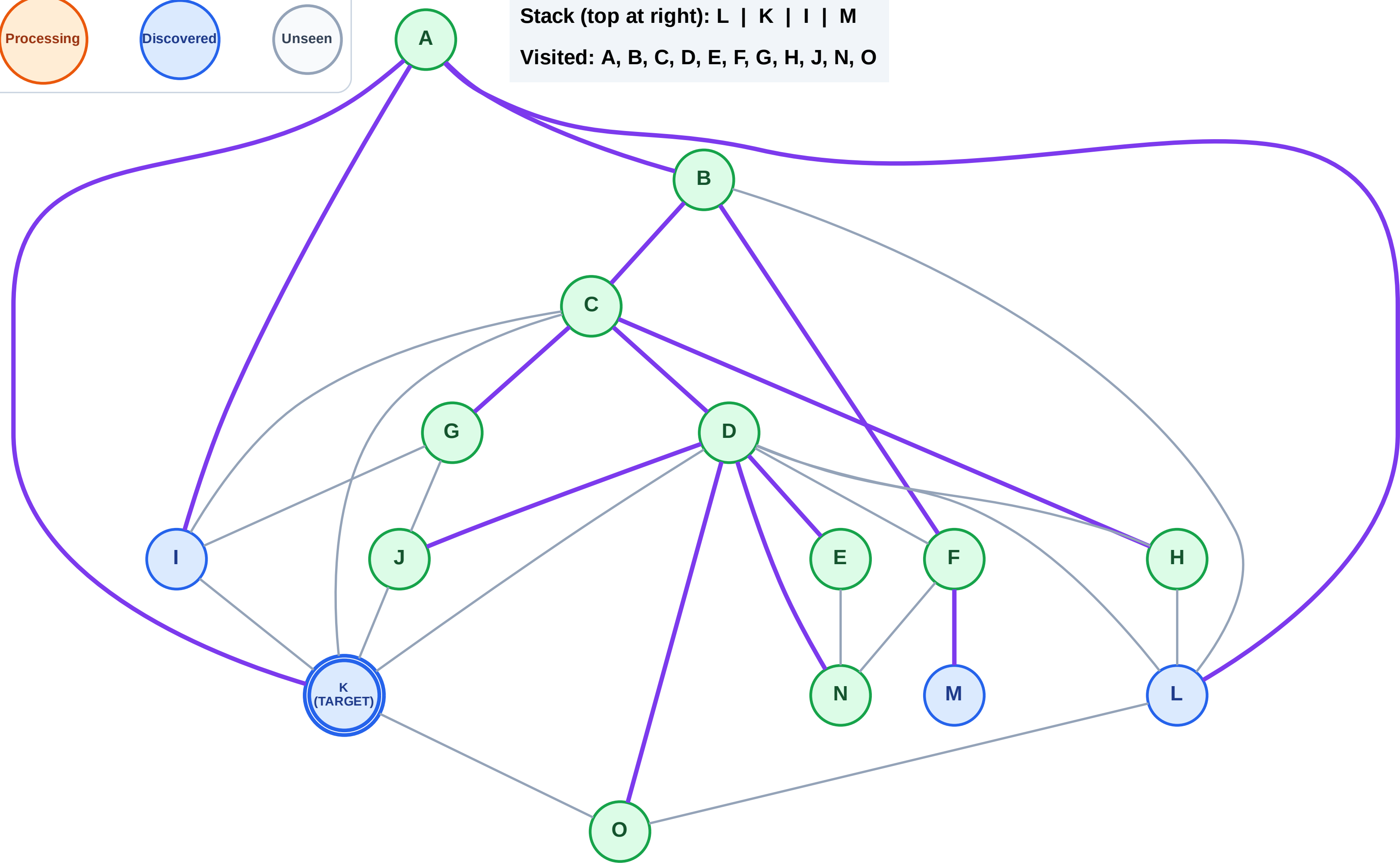
Legend

Complete

Processing

Discovered

Unseen



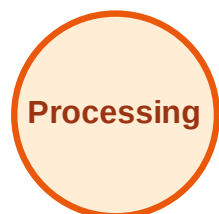
DFS Traversal

Step 24: Process node M

Legend



Complete



Processing



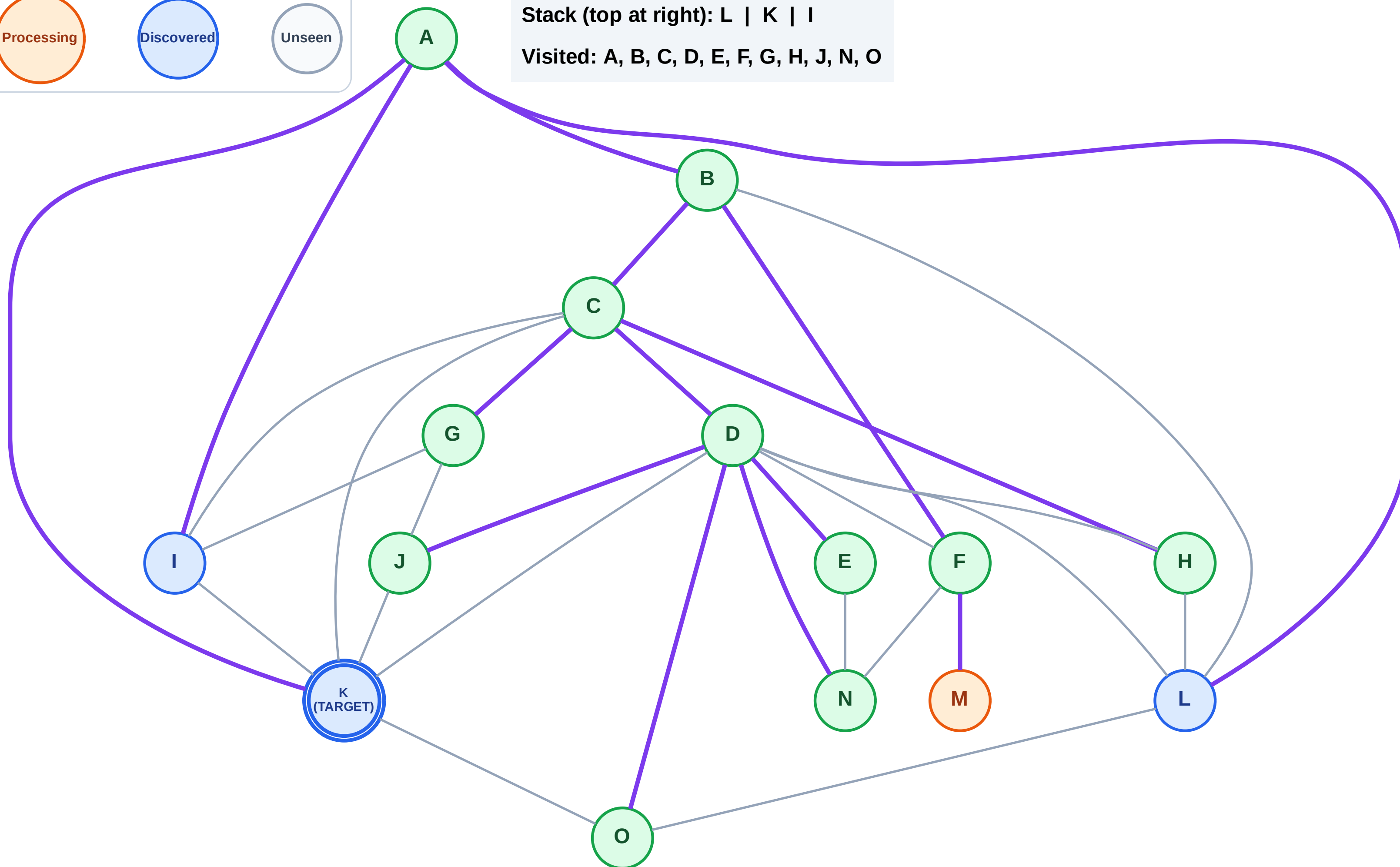
Discovered



Unseen

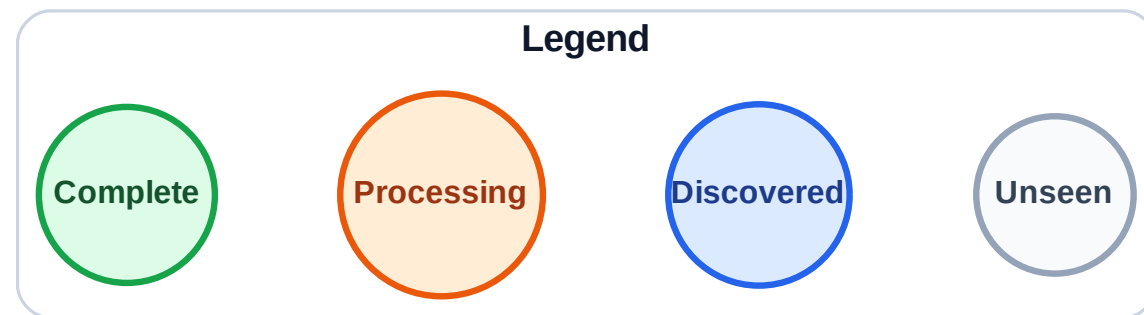
Stack (top at right): L | K | I

Visited: A, B, C, D, E, F, G, H, J, N, O



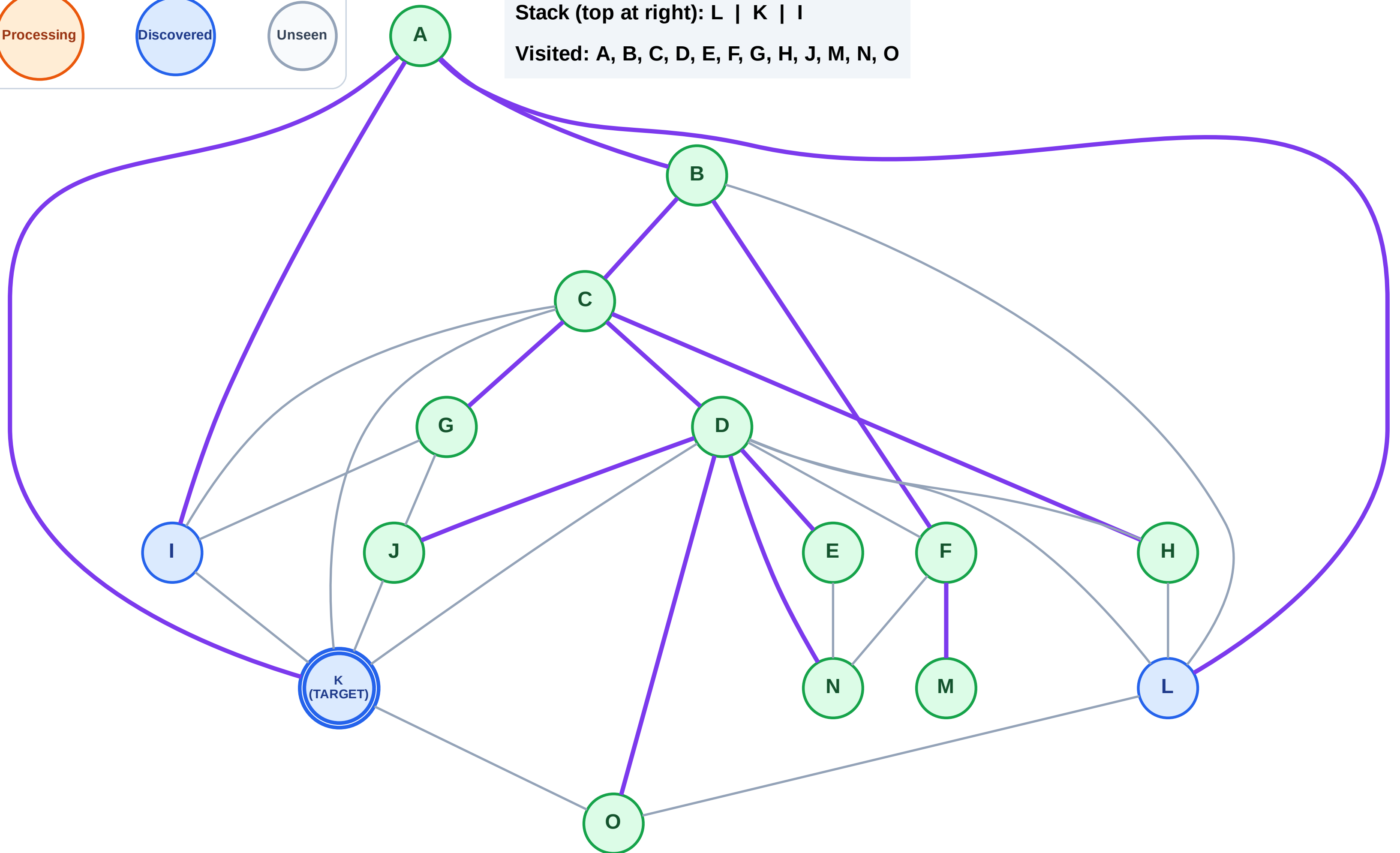
DFS Traversal

Step 25: No new neighbors from M



Stack (top at right): L | K | I

Visited: A, B, C, D, E, F, G, H, J, M, N, O



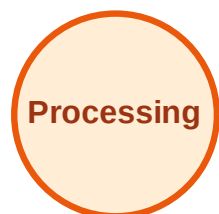
DFS Traversal

Step 26: Process node I

Legend



Complete



Processing



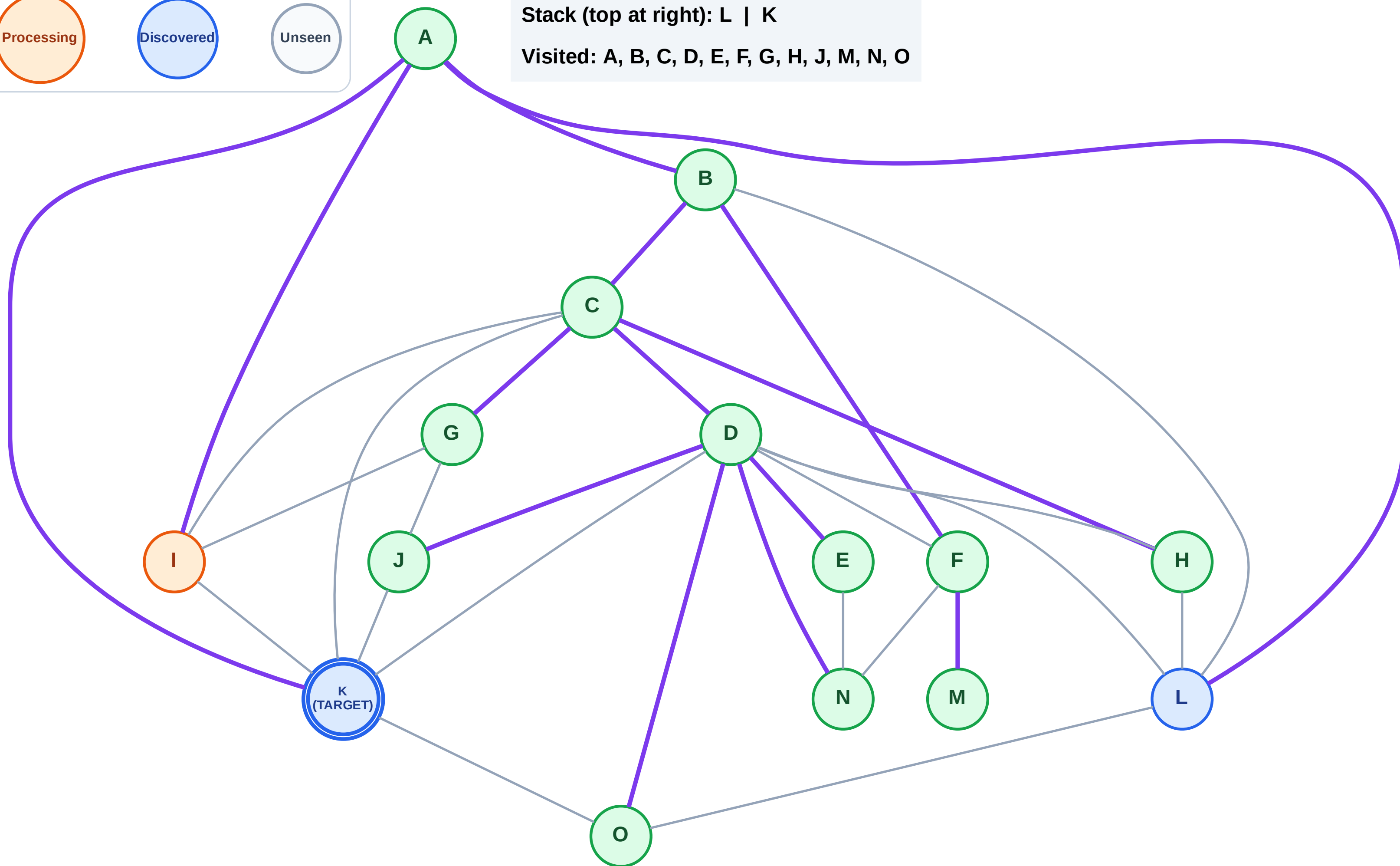
Discovered



Unseen

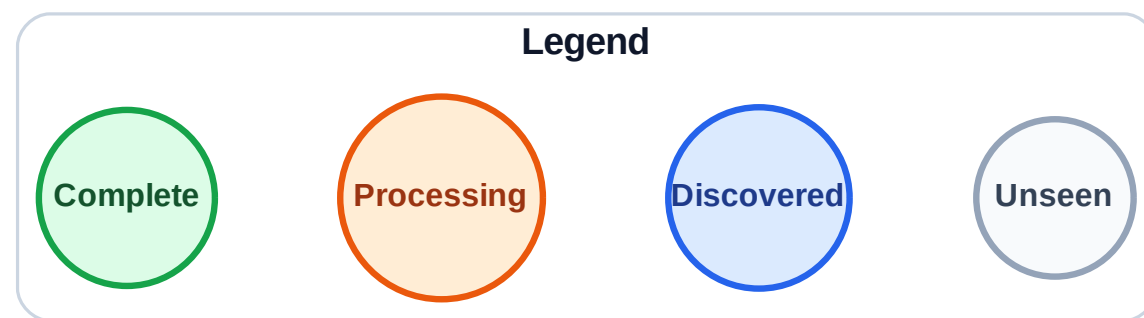
Stack (top at right): L | K

Visited: A, B, C, D, E, F, G, H, J, M, N, O



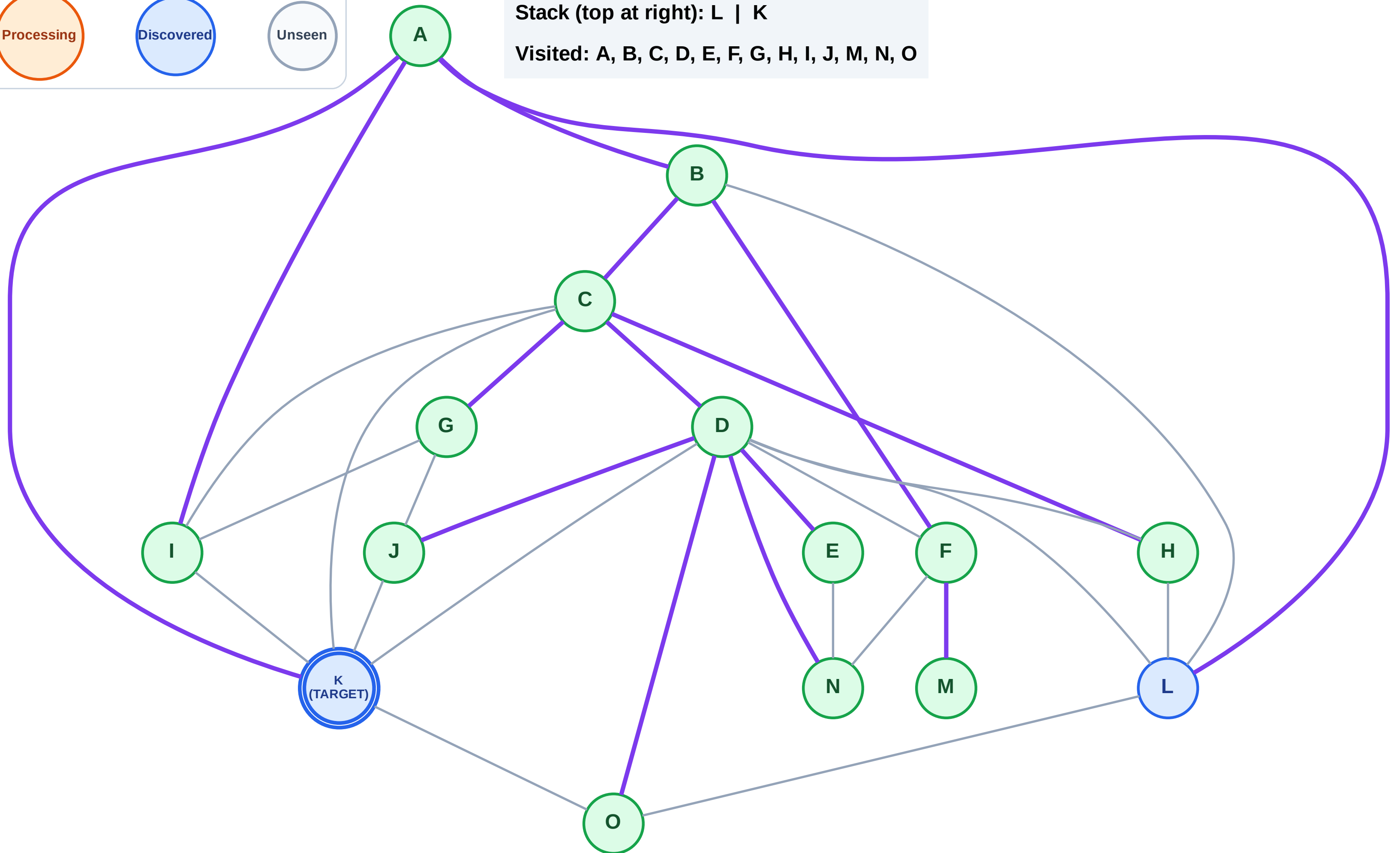
DFS Traversal

Step 27: No new neighbors from I



Stack (top at right): L | K

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



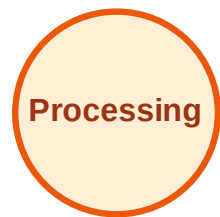
DFS Traversal

Step 28: Process node K

Legend



Complete



Processing



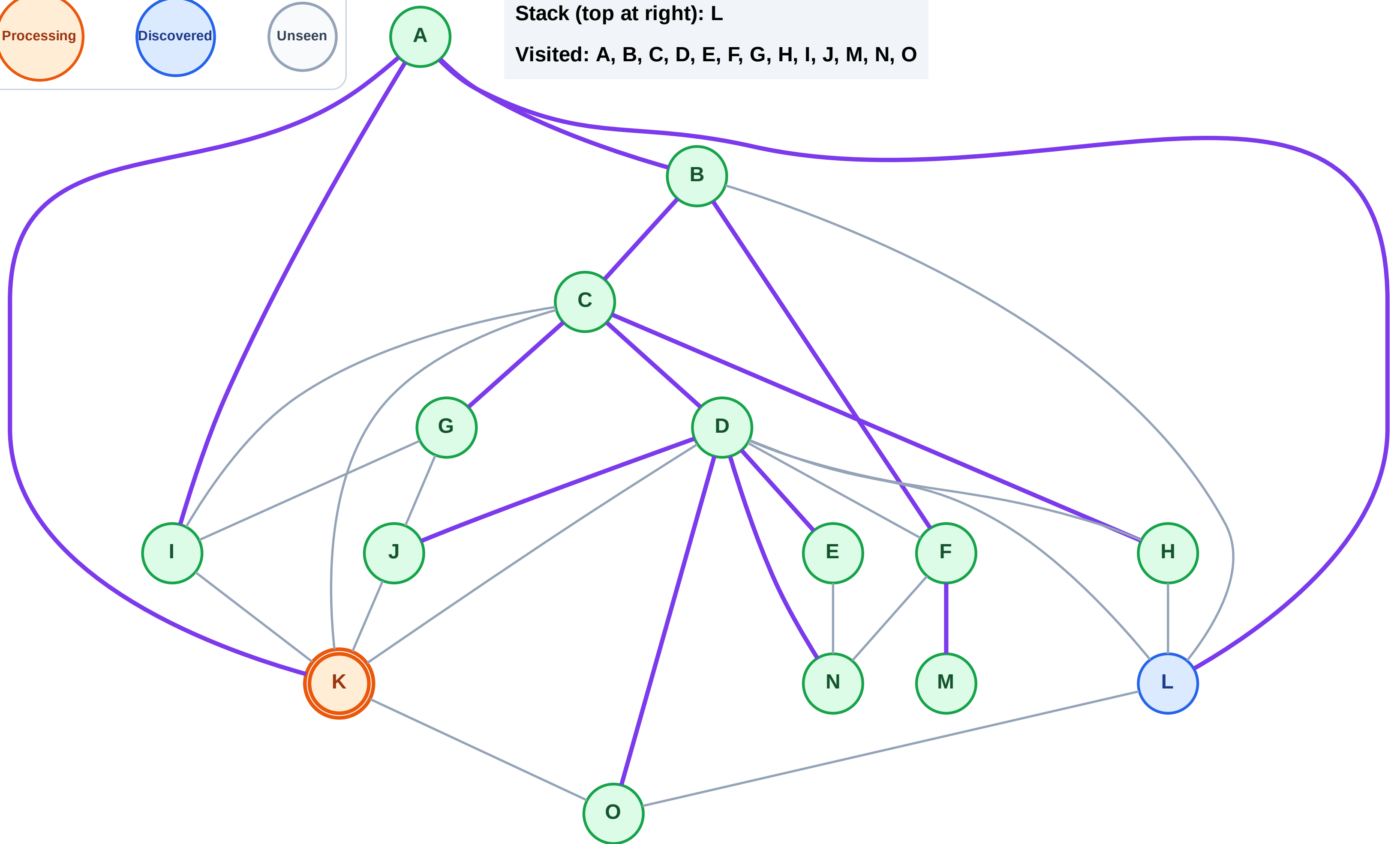
Discovered



Unseen

Stack (top at right): L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



DFS Traversal

Step 29: TARGET NODE K REACHED! Path (1 hops): A → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O

